A First Course in PROBABILITY

EIGHTH EDITION



SHELDON ROSS

the value of F is constant in the intervals $[x_{i-1}, x_i)$ and then takes a step (or jump) of size $p(x_i)$ at x_i. For instance, if X has a probability mass function given by

$$p(1) = \frac{1}{4} \qquad p(2) = \frac{1}{2} \qquad p(3) = \frac{1}{8} \qquad p(4) = \frac{1}{8}$$

then its cumulative distribution function is

$$F(a) = \begin{cases} 0 & a < 1 \\ \frac{1}{4} & 1 \le a < 2 \\ \frac{3}{4} & 2 \le a < 3 \\ \frac{7}{8} & 3 \le a < 4 \\ 1 & 4 \le a \end{cases}$$

This function is depicted graphically in Figure 4.3.

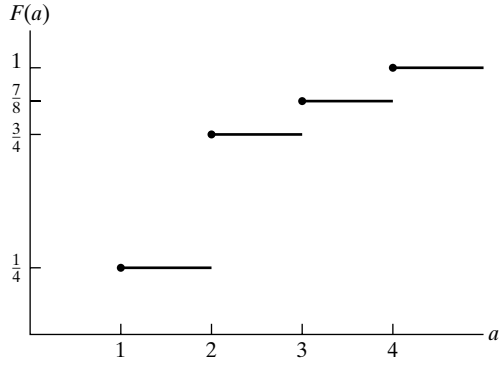


FIGURE 4.3

Note that the size of the step at any of the values 1, 2, 3, and 4 is equal to the probability that X assumes that particular value.

4.3 EXPECTED VALUE

One of the most important concepts in probability theory is that of the expectation of a random variable. If X is a discrete random variable having a probability mass function $p(x)$, then the *expectation*, or the *expected value*, of X, denoted by $E[X]$, is defined by

$$E[X] = \sum_{x:p(x)>0} xp(x)$$

In words, the expected value of X is a weighted average of the possible values that X can take on, each value being weighted by the probability that X assumes it. For instance, on the one hand, if the probability mass function of X is given by

$$p(0) = \frac{1}{2} = p(1)$$

then

$$E[X] = 0\left(\frac{1}{2}\right) + 1\left(\frac{1}{2}\right) = \frac{1}{2}$$

is just the ordinary average of the two possible values, 0 and 1, that X can assume. On the other hand, if

$$p(0) = \frac{1}{3} \qquad p(1) = \frac{2}{3}$$

then

$$E[X] = 0\left(\frac{1}{3}\right) + 1\left(\frac{2}{3}\right) = \frac{2}{3}$$

is a weighted average of the two possible values 0 and 1, where the value 1 is given twice as much weight as the value 0, since $p(1) = 2p(0)$.

Another motivation of the definition of expectation is provided by the frequency interpretation of probabilities. This interpretation (partially justified by the strong law of large numbers, to be presented in Chapter 8) assumes that if an infinite sequence of independent replications of an experiment is performed, then, for any event E, the proportion of time that E occurs will be $P(E)$. Now, consider a random variable X that must take on one of the values $x_1, x_2, \ldots x_n$ with respective probabilities $p(x_1), p(x_2), \ldots, p(x_n)$, and think of X as representing our winnings in a single game of chance. That is, with probability $p(x_i)$ we shall win x_i units $i = 1, 2, \ldots, n$. By the frequency interpretation, if we play this game continually, then the proportion of time that we win x_i will be $p(x_i)$. Since this is true for all $i, i = 1, 2, \ldots, n$, it follows that our average winnings per game will be

$$\sum_{i=1}^{n} x_i p(x_i) = E[X]$$

EXAMPLE 3a

Find $E[X]$, where X is the outcome when we roll a fair die.

Solution. Since $p(1) = p(2) = p(3) = p(4) = p(5) = p(6) = \frac{1}{6}$, we obtain

$$E[X] = 1\left(\frac{1}{6}\right) + 2\left(\frac{1}{6}\right) + 3\left(\frac{1}{6}\right) + 4\left(\frac{1}{6}\right) + 5\left(\frac{1}{6}\right) + 6\left(\frac{1}{6}\right) = \frac{7}{2}$$

■

EXAMPLE 3b

We say that I is an indicator variable for the event A if

$$I = \begin{cases} 1 & \text{if } A \text{ occurs} \\ 0 & \text{if } A^c \text{ occurs} \end{cases}$$

Find $E[I]$.

Solution. Since $p(1) = P(A), p(0) = 1 - P(A)$, we have

$$E[I] = P(A)$$

That is, the expected value of the indicator variable for the event A is equal to the probability that A occurs. ■

EXAMPLE 3c

A contestant on a quiz show is presented with two questions, questions 1 and 2, which he is to attempt to answer in some order he chooses. If he decides to try question i first, then he will be allowed to go on to question j, $j \neq i$, only if his answer to question i is correct. If his initial answer is incorrect, he is not allowed to answer the other question. The contestant is to receive V_i dollars if he answers question i correctly, $i = 1, 2$. For instance, he will receive $V_1 + V_2$ dollars if he answers both questions correctly. If the probability that he knows the answer to question i is $P_i, i = 1, 2$, which question should he attempt to answer first so as to maximize his expected winnings? Assume that the events $E_i, i = 1, 2$, that he knows the answer to question i are independent events.

Solution. On the one hand, if he attempts to answer question 1 first, then he will win

$$\begin{array}{ll} 0 & \text{with probability } 1 - P_1 \\ V_1 & \text{with probability } P_1(1 - P_2) \\ V_1 + V_2 & \text{with probability } P_1P_2 \end{array}$$

Hence, his expected winnings in this case will be

$$V_1P_1(1 - P_2) + (V_1 + V_2)P_1P_2$$

On the other hand, if he attempts to answer question 2 first, his expected winnings will be

$$V_2P_2(1 - P_1) + (V_1 + V_2)P_1P_2$$

Therefore, it is better to try question 1 first if

$$V_1P_1(1 - P_2) \geq V_2P_2(1 - P_1)$$

or, equivalently, if

$$\frac{V_1P_1}{1 - P_1} \geq \frac{V_2P_2}{1 - P_2}$$

For example, if he is 60 percent certain of answering question 1, worth \$200, correctly and he is 80 percent certain of answering question 2, worth \$100, correctly, then he should attempt to answer question 2 first because

$$400 = \frac{(100)(.8)}{.2} > \frac{(200)(.6)}{.4} = 300$$ ■

EXAMPLE 3d

A school class of 120 students is driven in 3 buses to a symphonic performance. There are 36 students in one of the buses, 40 in another, and 44 in the third bus. When the buses arrive, one of the 120 students is randomly chosen. Let X denote the number of students on the bus of that randomly chosen student, and find $E[X]$.

Solution. Since the randomly chosen student is equally likely to be any of the 120 students, it follows that

$$P\{X = 36\} = \frac{36}{120} \quad P\{X = 40\} = \frac{40}{120} \quad P\{X = 44\} = \frac{44}{120}$$

Hence,

$$E[X] = 36\left(\frac{3}{10}\right) + 40\left(\frac{1}{3}\right) + 44\left(\frac{11}{30}\right) = \frac{1208}{30} = 40.2667$$

However, the average number of students on a bus is $120/3 = 40$, showing that the expected number of students on the bus of a randomly chosen student is larger than the average number of students on a bus. This is a general phenomenon, and it occurs because the more students there are on a bus, the more likely it is that a randomly chosen student would have been on that bus. As a result, buses with many students are given more weight than those with fewer students. (See Self-Test Problem 4.) ■

Remark. The probability concept of expectation is analogous to the physical concept of the *center of gravity* of a distribution of mass. Consider a discrete random variable X having probability mass function $p(x_i), i \geq 1$. If we now imagine a weightless rod in which weights with mass $p(x_i), i \geq 1$, are located at the points $x_i, i \geq 1$ (see Figure 4.4), then the point at which the rod would be in balance is known as the center of gravity. For those readers acquainted with elementary statics, it is now a simple matter to show that this point is at $E[X]$.† ■

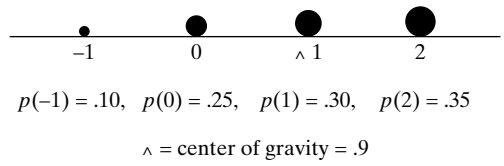


FIGURE 4.4

4.4 EXPECTATION OF A FUNCTION OF A RANDOM VARIABLE

Suppose that we are given a discrete random variable along with its probability mass function and that we want to compute the expected value of some function of X, say, $g(X)$. How can we accomplish this? One way is as follows: Since $g(X)$ is itself a discrete random variable, it has a probability mass function, which can be determined from the probability mass function of X. Once we have determined the probability mass function of $g(X)$, we can compute $E[g(X)]$ by using the definition of expected value.

EXAMPLE 4a

Let X denote a random variable that takes on any of the values -1, 0, and 1 with respective probabilities

$$P\{X = -1\} = .2 \quad P\{X = 0\} = .5 \quad P\{X = 1\} = .3$$

Compute $E[X^2]$.

Solution. Let $Y = X^2$. Then the probability mass function of Y is given by

$$P\{Y = 1\} = P\{X = -1\} + P\{X = 1\} = .5$$
$$P\{Y = 0\} = P\{X = 0\} = .5$$

Hence,

$$E[X^2] = E[Y] = 1(.5) + 0(.5) = .5$$

†To prove this, we must show that the sum of the torques tending to turn the point around $E[X]$ is equal to 0. That is, we must show that $0 = \sum_i (x_i - E[X])p(x_i)$, which is immediate.

Note that

$$.5 = E[X^2] \neq (E[X])^2 = .01$$ ■

Although the preceding procedure will always enable us to compute the expected value of any function of X from a knowledge of the probability mass function of X, there is another way of thinking about $E[g(X)]$: Since $g(X)$ will equal $g(x)$ whenever X is equal to x, it seems reasonable that $E[g(X)]$ should just be a weighted average of the values $g(x)$, with $g(x)$ being weighted by the probability that X is equal to x. That is, the following result is quite intuitive:

Proposition 4.1.
If X is a discrete random variable that takes on one of the values $x_i, i \geq 1$, with respective probabilities $p(x_i)$, then, for any real-valued function g,

$$E[g(X)] = \sum_i g(x_i)p(x_i)$$

Before proving this proposition, let us check that it is in accord with the results of Example 4a. Applying it to that example yields

$$\begin{aligned} E\{X^2\} &= (-1)^2(.2) + 0^2(.5) + 1^2(.3) \\ &= 1(.2 + .3) + 0(.5) \\ &= .5 \end{aligned}$$

which is in agreement with the result given in Example 4a.

Proof of Proposition 4.1: The proof of Proposition 4.1 proceeds, as in the preceding verification, by grouping together all the terms in $\sum_i g(x_i)p(x_i)$ having the same value of $g(x_i)$. Specifically, suppose that $y_j, j \geq 1$, represent the different values of $g(x_i), i \geq 1$. Then, grouping all the $g(x_i)$ having the same value gives

$$\begin{aligned} \sum_i g(x_i)p(x_i) &= \sum_j \sum_{i:g(x_i)=y_j} g(x_i)p(x_i) \\ &= \sum_j \sum_{i:g(x_i)=y_j} y_j p(x_i) \\ &= \sum_j y_j \sum_{i:g(x_i)=y_j} p(x_i) \\ &= \sum_j y_j P\{g(X) = y_j\} \\ &= E[g(X)] \end{aligned}$$ □

EXAMPLE 4b

A product that is sold seasonally yields a net profit of b dollars for each unit sold and a net loss of ℓ dollars for each unit left unsold when the season ends. The number of units of the product that are ordered at a specific department store during any season is a random variable having probability mass function $p(i), i \geq 0$. If the store must stock this product in advance, determine the number of units the store should stock so as to maximize its expected profit.

Solution. Let X denote the number of units ordered. If s units are stocked, then the profit—call it $P(s)$—can be expressed as

$$\begin{aligned} P(s) &= bX - (s - X)\ell \quad \text{if } X \le s \\ &= sb \qquad\qquad\qquad\quad \text{if } X > s \end{aligned}$$

Hence, the expected profit equals

$$\begin{aligned} E[P(s)] &= \sum_{i=0}^{s} [bi - (s - i)\ell]p(i) + \sum_{i=s+1}^{\infty} sbp(i) \\ &= (b + \ell)\sum_{i=0}^{s} ip(i) - s\ell\sum_{i=0}^{s} p(i) + sb\left[1 - \sum_{i=0}^{s} p(i)\right] \\ &= (b + \ell)\sum_{i=0}^{s} ip(i) - (b + \ell)s\sum_{i=0}^{s} p(i) + sb \\ &= sb + (b + \ell)\sum_{i=0}^{s} (i - s)p(i) \end{aligned}$$

To determine the optimum value of s, let us investigate what happens to the profit when we increase s by 1 unit. By substitution, we see that the expected profit in this case is given by

$$\begin{aligned} E[P(s + 1)] &= b(s + 1) + (b + \ell)\sum_{i=0}^{s+1} (i - s - 1)p(i) \\ &= b(s + 1) + (b + \ell)\sum_{i=0}^{s} (i - s - 1)p(i) \end{aligned}$$

Therefore,

$$E[P(s + 1)] - E[P(s)] = b - (b + \ell)\sum_{i=0}^{s} p(i)$$

Thus, stocking $s + 1$ units will be better than stocking s units whenever

$$\sum_{i=0}^{s} p(i) < \frac{b}{b + \ell} \tag{4.1}$$

Because the left-hand side of Equation (4.1) is increasing in s while the right-hand side is constant, the inequality will be satisfied for all values of $s \le s^*$, where s^* is the largest value of s satisfying Equation (4.1). Since

$$E[P(0)] < \cdots < E[P(s^*)] < E[P(s^* + 1)] > E[P(s^* + 2)] > \cdots$$

it follows that stocking $s^* + 1$ items will lead to a maximum expected profit. ■

EXAMPLE 4c Utility

Suppose that you must choose one of two possible actions, each of which can result in any of n consequences, denoted as $C_1, \ldots, C_n$. Suppose that if the first action is

chosen, then consequence C_i will result with probability $p_i, i = 1, \ldots, n$, whereas if the second action is chosen, then consequence C_i will result with probability q_i, $i = 1, \ldots, n$, where $\sum_{i=1}^{n} p_i = \sum_{i=1}^{n} q_i = 1$. The following approach can be used to determine which action to choose: Start by assigning numerical values to the different consequences in the following manner: First, identify the least and the most desirable consequences—call them c and C, respectively; give consequence c the value 0 and give C the value 1. Now consider any of the other $n - 2$ consequences, say, C_i. To value this consequence, imagine that you are given the choice between either receiving C_i or taking part in a random experiment that either earns you consequence C with probability u or consequence c with probability $1 - u$. Clearly, your choice will depend on the value of u. On the one hand, if $u = 1$, then the experiment is certain to result in consequence C, and since C is the most desirable consequence, you will prefer participating in the experiment to receiving C_i. On the other hand, if $u = 0$, then the experiment will result in the least desirable consequence—namely, c—so in this case you will prefer the consequence C_i to participating in the experiment. Now, as u decreases from 1 to 0, it seems reasonable that your choice will at some point switch from participating in the experiment to the certain return of C_i, and at that critical switch point you will be indifferent between the two alternatives. Take that indifference probability u as the value of the consequence C_i. In other words, the value of C_i is that probability u such that you are indifferent between either receiving the consequence C_i or taking part in an experiment that returns consequence C with probability u or consequence c with probability $1 - u$. We call this indifference probability the *utility* of the consequence C_i, and we designate it as $u(C_i)$.

To determine which action is superior, we need to evaluate each one. Consider the first action, which results in consequence C_i with probability $p_i, i = 1, \ldots, n$. We can think of the result of this action as being determined by a two-stage experiment. In the first stage, one of the values $1, \ldots, n$ is chosen according to the probabilities $p_1, \ldots, p_n$; if value i is chosen, you receive consequence C_i. However, since C_i is equivalent to obtaining consequence C with probability $u(C_i)$ or consequence c with probability $1 - u(C_i)$, it follows that the result of the two-stage experiment is equivalent to an experiment in which either consequence C or consequence c is obtained, with C being obtained with probability

$$\sum_{i=1}^{n} p_i u(C_i)$$

Similarly, the result of choosing the second action is equivalent to taking part in an experiment in which either consequence C or consequence c is obtained, with C being obtained with probability

$$\sum_{i=1}^{n} q_i u(C_i)$$

Since C is preferable to c, it follows that the first action is preferable to the second action if

$$\sum_{i=1}^{n} p_i u(C_i) > \sum_{i=1}^{n} q_i u(C_i)$$

In other words, the worth of an action can be measured by the expected value of the utility of its consequence, and the action with the largest expected utility is the most preferable. ■

A simple logical consequence of Proposition 4.1 is Corollary 4.1.

Corollary 4.1. If a and b are constants, then

$$E[aX + b] = aE[X] + b$$

Proof.

$$\begin{aligned} E[aX + b] &= \sum_{x:p(x)>0} (ax + b)p(x) \\ &= a \sum_{x:p(x)>0} xp(x) + b \sum_{x:p(x)>0} p(x) \\ &= aE[X] + b \end{aligned}$$

□

The expected value of a random variable X, $E[X]$, is also referred to as the *mean* or the *first moment of X*. The quantity $E[X^n], n \geq 1$, is called the *nth moment of X*. By Proposition 4.1, we note that

$$E[X^n] = \sum_{x:p(x)>0} x^n p(x)$$

4.5 VARIANCE

Given a random variable X along with its distribution function F, it would be extremely useful if we were able to summarize the essential properties of F by certain suitably defined measures. One such measure would be $E[X]$, the expected value of X. However, although $E[X]$ yields the weighted average of the possible values of X, it does not tell us anything about the variation, or spread, of these values. For instance, although random variables W, Y, and Z having probability mass functions determined by

$$W = 0 \quad \text{with probability 1}$$

$$Y = \begin{cases} -1 & \text{with probability } \frac{1}{2} \\ +1 & \text{with probability } \frac{1}{2} \end{cases}$$

$$Z = \begin{cases} -100 & \text{with probability } \frac{1}{2} \\ +100 & \text{with probability } \frac{1}{2} \end{cases}$$

all have the same expectation—namely, 0—there is a much greater spread in the possible values of Y than in those of W (which is a constant) and in the possible values of Z than in those of Y.

Because we expect X to take on values around its mean $E[X]$, it would appear that a reasonable way of measuring the possible variation of X would be to look at how far apart X would be from its mean, on the average. One possible way to measure this variation would be to consider the quantity $E[|X - \mu|]$, where $\mu = E[X]$. However, it turns out to be mathematically inconvenient to deal with this quantity, so a more tractable quantity is usually considered—namely, the expectation of the square of the difference between X and its mean. We thus have the following definition.

Definition

If X is a random variable with mean μ, then the variance of X, denoted by $\text{Var}(X)$, is defined by

$$\text{Var}(X) = E[(X - \mu)^2]$$

An alternative formula for $\text{Var}(X)$ is derived as follows:

$$\begin{aligned}
\text{Var}(X) &= E[(X - \mu)^2] \\
&= \sum_x (x - \mu)^2 p(x) \\
&= \sum_x (x^2 - 2\mu x + \mu^2) p(x) \\
&= \sum_x x^2 p(x) - 2\mu \sum_x x p(x) + \mu^2 \sum_x p(x) \\
&= E[X^2] - 2\mu^2 + \mu^2 \\
&= E[X^2] - \mu^2
\end{aligned}$$

That is,

$$\boxed{\text{Var}(X) = E[X^2] - (E[X])^2}$$

In words, the variance of X is equal to the expected value of X^2 minus the square of its expected value. In practice, this formula frequently offers the easiest way to compute $\text{Var}(X)$.

EXAMPLE 5a

Calculate $\text{Var}(X)$ if X represents the outcome when a fair die is rolled.

Solution. It was shown in Example 3a that $E[X] = \frac{7}{2}$. Also,

$$\begin{aligned}
E[X^2] &= 1^2\left(\frac{1}{6}\right) + 2^2\left(\frac{1}{6}\right) + 3^2\left(\frac{1}{6}\right) + 4^2\left(\frac{1}{6}\right) + 5^2\left(\frac{1}{6}\right) + 6^2\left(\frac{1}{6}\right) \\
&= \left(\frac{1}{6}\right)(91)
\end{aligned}$$

Hence,

$$\text{Var}(X) = \frac{91}{6} - \left(\frac{7}{2}\right)^2 = \frac{35}{12}$$

■

A useful identity is that, for any constants a and b,

$$\text{Var}(aX + b) = a^2 \text{Var}(X)$$

To prove this equality, let $\mu = E[X]$ and note from Corollary 4.1 that $E[aX + b] = a\mu + b$. Therefore,

$$\begin{aligned}\text{Var}(aX + b) &= E[(aX + b - a\mu - b)^2] \\ &= E[a^2(X - \mu)^2] \\ &= a^2E[(X - \mu)^2] \\ &= a^2\text{Var}(X)\end{aligned}$$

Remarks. (a) Analogous to the means being the center of gravity of a distribution of mass, the variance represents, in the terminology of mechanics, the moment of inertia.

(b) The square root of the $\text{Var}(X)$ is called the *standard deviation* of X, and we denote it by $\text{SD}(X)$. That is,

$$\text{SD}(X) = \sqrt{\text{Var}(X)}$$

Discrete random variables are often classified according to their probability mass functions. In the next few sections, we consider some of the more common types.

4.6 THE BERNOULLI AND BINOMIAL RANDOM VARIABLES

Suppose that a trial, or an experiment, whose outcome can be classified as either a *success* or a *failure* is performed. If we let $X = 1$ when the outcome is a success and $X = 0$ when it is a failure, then the probability mass function of X is given by

$$\begin{aligned}p(0) &= P\{X = 0\} = 1 - p \\ p(1) &= P\{X = 1\} = p\end{aligned} \tag{6.1}$$

where $p, 0 \le p \le 1$, is the probability that the trial is a success.

A random variable X is said to be a *Bernoulli random variable* (after the Swiss mathematician James Bernoulli) if its probability mass function is given by Equations (6.1) for some $p \in (0, 1)$.

Suppose now that n independent trials, each of which results in a success with probability p and in a failure with probability $1 - p$, are to be performed. If X represents the number of successes that occur in the n trials, then X is said to be a *binomial random variable* with parameters (n, p). Thus, a Bernoulli random variable is just a binomial random variable with parameters $(1, p)$.

The probability mass function of a binomial random variable having parameters (n, p) is given by

$$p(i) = \binom{n}{i} p^i(1 - p)^{n-i} \qquad i = 0, 1, \ldots, n \tag{6.2}$$

The validity of Equation (6.2) may be verified by first noting that the probability of any particular sequence of n outcomes containing i successes and $n - i$ failures is, by the assumed independence of trials, $p^i(1 - p)^{n-i}$. Equation (6.2) then follows, since there are $\binom{n}{i}$ different sequences of the n outcomes leading to i successes and $n - i$ failures. This perhaps can most easily be seen by noting that there are $\binom{n}{i}$ different choices of the i trials that result in successes. For instance, if $n = 4, i = 2$, then there are $\binom{4}{2} = 6$ ways in which the four trials can result in two successes,

The zeta distribution owes its name to the fact that the function

$$\zeta(s) = 1 + \left(\frac{1}{2}\right)^s + \left(\frac{1}{3}\right)^s + \cdots + \left(\frac{1}{k}\right)^s + \cdots$$

is known in mathematical disciplines as the Riemann zeta function (after the German mathematician G. F. B. Riemann).

The zeta distribution was used by the Italian economist V. Pareto to describe the distribution of family incomes in a given country. However, it was G. K. Zipf who applied zeta distribution to a wide variety of problems in different areas and, in doing so, popularized its use.

4.9 EXPECTED VALUE OF SUMS OF RANDOM VARIABLES

A very important property of expectations is that the expected value of a sum of random variables is equal to the sum of their expectations. In this section, we will prove this result under the assumption that the set of possible values of the probability experiment—that is, the sample space S—is either finite or countably infinite. Although the result is true without this assumption (and a proof is outlined in the theoretical exercises), not only will the assumption simplify the argument, but it will also result in an enlightening proof that will add to our intuition about expectations. So, for the remainder of this section, suppose that the sample space S is either a finite or a countably infinite set.

For a random variable X, let $X(s)$ denote the value of X when $s \in S$ is the outcome of the experiment. Now, if X and Y are both random variables, then so is their sum. That is, $Z = X + Y$ is also a random variable. Moreover, $Z(s) = X(s) + Y(s)$.

EXAMPLE 9a

Suppose that the experiment consists of flipping a coin 5 times, with the outcome being the resulting sequence of heads and tails. Suppose X is the number of heads in the first 3 flips and Y is the number of heads in the final 2 flips. Let $Z = X + Y$. Then, for instance, for the outcome $s = (h, t, h, t, h)$,

$$\begin{aligned} X(s) &= 2 \\ Y(s) &= 1 \\ Z(s) &= X(s) + Y(s) = 3 \end{aligned}$$

meaning that the outcome (h, t, h, t, h) results in 2 heads in the first three flips, 1 head in the final two flips, and a total of 3 heads in the five flips. ■

Let $p(s) = P(\{s\})$ be the probability that s is the outcome of the experiment. Because we can write any event A as the finite or countably infinite union of the mutually exclusive events $\{s\}, s \in A$, it follows by the axioms of probability that

$$P(A) = \sum_{s \in A} p(s)$$

When $A = S$, the preceding equation gives

$$1 = \sum_{s \in S} p(s)$$

Now, let X be a random variable, and consider $E[X]$. Because $X(s)$ is the value of X when s is the outcome of the experiment, it seems intuitive that $E[X]$—the weighted average of the possible values of X, with each value weighted by the probability that X assumes that value—should equal a weighted average of the values $X(s), s \in S$, with $X(s)$ weighted by the probability that s is the outcome of the experiment. We now prove this intuition.

Proposition 9.1.

$$E[X] = \sum_{s \in S} X(s)\, p(s)$$

Proof. Suppose that the distinct values of X are $x_i, i \geq 1$. For each i, let S_i be the event that X is equal to x_i. That is, $S_i = \{s : X(s) = x_i\}$. Then,

$$\begin{aligned} E[X] &= \sum_i x_i P\{X = x_i\} \\ &= \sum_i x_i P(S_i) \\ &= \sum_i x_i \sum_{s \in S_i} p(s) \\ &= \sum_i \sum_{s \in S_i} x_i p(s) \\ &= \sum_i \sum_{s \in S_i} X(s) p(s) \\ &= \sum_{s \in S} X(s) p(s) \end{aligned}$$

where the final equality follows because $S_1, S_2, \ldots$ are mutually exclusive events whose union is S. □

EXAMPLE 9b

Suppose that two independent flips of a coin that comes up heads with probability p are made, and let X denote the number of heads obtained. Because

$$\begin{aligned} P(X = 0) &= P(t,t) = (1 - p)^2, \\ P(X = 1) &= P(h,t) + P(t,h) = 2p(1 - p) \\ P(X = 2) &= P(h,h) = p^2 \end{aligned}$$

it follows from the definition of expected value that

$$E[X] = 0 \cdot (1 - p)^2 + 1 \cdot 2p(1 - p) + 2 \cdot p^2 = 2p$$

which agrees with

$$\begin{aligned} E[X] &= X(h,h)p^2 + X(h,t)p(1 - p) + X(t,h)(1 - p)p + X(t,t)(1 - p)^2 \\ &= 2p^2 + p(1 - p) + (1 - p)p \\ &= 2p \end{aligned}$$

■

We now prove the important and useful result that the expected value of a sum of random variables is equal to the sum of their expectations.

Corollary 9.2. For random variables $X_1, X_2, \ldots, X_n$,

$$E\left[\sum_{i=1}^{n} X_i\right] = \sum_{i=1}^{n} E[X_i]$$

Proof. Let $Z = \sum_{i=1}^{n} X_i$. Then, by Proposition 9.1,

$$\begin{aligned} E[Z] &= \sum_{s \in S} Z(s)p(s) \\ &= \sum_{s \in S} \big(X_1(s) + X_2(s) + \ldots + X_n(s)\big)\, p(s) \\ &= \sum_{s \in S} X_1(s)p(s) + \sum_{s \in S} X_2(s)p(s) + \ldots + \sum_{s \in S} X_n(s)p(s) \\ &= E[X_1] + E[X_2] + \ldots + E[X_n] \end{aligned}$$

■

EXAMPLE 9c

Find the expected value of the sum obtained when n fair dice are rolled.

Solution. Let X be the sum. We will compute $E[X]$ by using the representation

$$X = \sum_{i=1}^{n} X_i$$

where X_i is the upturned value on die i. Because X_i is equally likely to be any of the values from 1 to 6, it follows that

$$E[X_i] = \sum_{i=1}^{6} i(1/6) = 21/6 = 7/2$$

which yields the result

$$E[X] = E\left[\sum_{i=1}^{n} X_i\right] = \sum_{i=1}^{n} E[X_i] = 3.5\, n$$

■

EXAMPLE 9d

Find the expected total number of successes that result from n trials when trial i is a success with probability p_i, $i = 1, \ldots, n$.

Solution. Letting

$$X_i = \begin{cases} 1, & \text{if trial } i \text{ is a success} \\ 0, & \text{if trial } i \text{ is a failure} \end{cases}$$

we have the representation

$$X = \sum_{i=1}^{n} X_i$$

Consequently,

$$E[X] = \sum_{i=1}^{n} E[X_i] = \sum_{i=1}^{n} p_i$$

Note that this result does not require that the trials be independent. It includes as a special case the expected value of a binomial random variable, which assumes independent trials and all $p_i = p$, and thus has mean np. It also gives the expected value of a hypergeometric random variable representing the number of white balls selected when n balls are randomly selected, without replacement, from an urn of N balls of which m are white. We can interpret the hypergeometric as representing the number of successes in n trials, where trial i is said to be a success if the ith ball selected is white. Because the ith ball selected is equally likely to be any of the N balls and thus has probability m/N of being white, it follows that the hypergeometric is the number of successes in n trials in which each trial is a success with probability $p = m/N$. Hence, even though these hypergeometric trials are dependent, it follows from the result of Example 9d that the expected value of the hypergeometric is $np = nm/N$. ■

EXAMPLE 9e

Derive an expression for the variance of the number of successful trials in Example 9*d*, and apply it to obtain the variance of a binomial random variable with parameters n and p, and of a hypergeometric random variable equal to the number of white balls chosen when n balls are randomly chosen from an urn containing N balls of which m are white.

Solution. Letting X be the number of successful trials, and using the same representation for X—namely, $X = \sum_{i=1}^{n} X_i$—as in the previous example, we have

$$\begin{aligned}
E[X^2] &= E\left[\left(\sum_{i=1}^{n} X_i\right)\left(\sum_{j=1}^{n} X_j\right)\right] \\
&= E\left[\sum_{i=1}^{n} X_i\left(X_i + \sum_{j\neq i} X_j\right)\right] \\
&= E\left[\sum_{i=1}^{n} X_i^2 + \sum_{i=1}^{n}\sum_{j\neq i} X_i X_j\right] \\
&= \sum_{i=1}^{n} E[X_i^2] + \sum_{i=1}^{n}\sum_{j\neq i} E[X_i X_j] \\
&= \sum_{i} p_i + \sum_{i=1}^{n}\sum_{j\neq i} E[X_i X_j] \qquad (9.1)
\end{aligned}$$

where the final equation used that $X_i^2 = X_i$. However, because the possible values of both X_i and X_j are 0 or 1, it follows that

$$X_i X_j = \begin{cases} 1, & \text{if } X_i = 1, X_j = 1 \\ 0, & \text{otherwise} \end{cases}$$

Hence,

$$E[X_iX_j] = P\{X_i = 1, X_j = 1\} = P(\text{trials } i \text{ and } j \text{ are successes})$$

Now, on the one hand, if X is binomial, then, for $i \neq j$, the results of trial i and trial j are independent, with each being a success with probability p. Therefore,

$$E[X_iX_j] = p^2, \qquad i \neq j$$

Together with Equation (9.1), the preceding equation shows that, for a binomial random variable X,

$$E[X^2] = np + n(n - 1)p^2$$

implying that

$$\text{Var}(X) = E[X^2] - (E[X])^2 = np + n(n - 1)p^2 - n^2p^2 = np(1 - p)$$

On the other hand, if X is hypergeometric, then, given that a white ball is chosen in trial i, each of the other $N - 1$ balls, of which $m - 1$ are white, is equally likely to be the jth ball chosen, for $j \neq i$. Consequently, for $j \neq i$,

$$P\{X_i = 1, X_j = 1\} = P\{X_i = 1\}P\{X_j = 1|X_i = 1\} = \frac{m}{N}\frac{m - 1}{N - 1}$$

Using $p_i = m/N$, we now obtain, from Equation (9.1),

$$E[X^2] = \frac{nm}{N} + n(n - 1)\frac{m}{N}\frac{m - 1}{N - 1}$$

Consequently,

$$\text{Var}(X) = \frac{nm}{N} + n(n - 1)\frac{m}{N}\frac{m - 1}{N - 1} - \left(\frac{nm}{N}\right)^2$$

which, as shown in Example 8j, can be simplified to yield

$$\text{Var}(X) = np(1 - p)\left(1 - \frac{n - 1}{N - 1}\right)$$

where $p = m/N$. ■

4.10 PROPERTIES OF THE CUMULATIVE DISTRIBUTION FUNCTION

Recall that, for the distribution function F of X, $F(b)$ denotes the probability that the random variable X takes on a value that is less than or equal to b. Following are some properties of the cumulative distribution function (c.d.f.) F:

1. F is a nondecreasing function; that is, if $a < b$, then $F(a) \leq F(b)$.
2. $\lim_{b\to\infty} F(b) = 1$.
3. $\lim_{b\to-\infty} F(b) = 0$.
4. F is right continuous. That is, for any b and any decreasing sequence $b_n, n \geq 1$, that converges to b, $\lim_{n\to\infty} F(b_n) = F(b)$.

Property 1 follows, as was noted in Section 4.1, because, for $a < b$, the event $\{X \leq a\}$ is contained in the event $\{X \leq b\}$ and so cannot have a larger probability. Properties 2, 3, and 4 all follow from the continuity property of probabilities

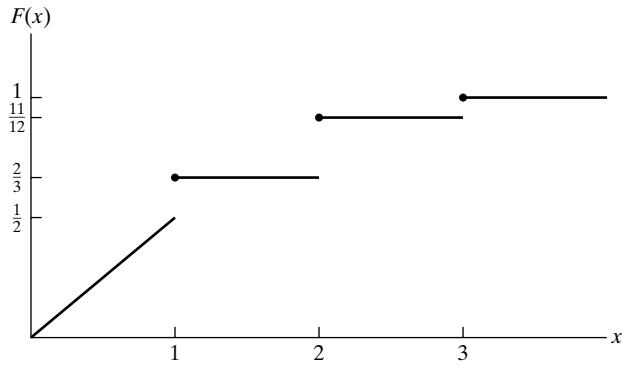


FIGURE 4.8: Graph of $F(x)$.

A graph of $F(x)$ is presented in Figure 4.8. Compute (a) $P\{X < 3\}$, (b) $P\{X = 1\}$, (c) $P\{X > \frac{1}{2}\}$, and (d) $P\{2 < X \leq 4\}$.

Solution. (a) $P\{X < 3\} = \lim_n P\left\{X \leq 3 - \frac{1}{n}\right\} = \lim_n F\left(3 - \frac{1}{n}\right) = \frac{11}{12}$

(b)

$$\begin{aligned} P\{X = 1\} &= P\{X \leq 1\} - P\{X < 1\} \\ &= F(1) - \lim_n F\left(1 - \frac{1}{n}\right) = \frac{2}{3} - \frac{1}{2} = \frac{1}{6} \end{aligned}$$

(c)

$$\begin{aligned} P\left\{X > \frac{1}{2}\right\} &= 1 - P\left\{X \leq \frac{1}{2}\right\} \\ &= 1 - F\left(\frac{1}{2}\right) = \frac{3}{4} \end{aligned}$$

(d)

$$\begin{aligned} P\{2 < X \leq 4\} &= F(4) - F(2) \\ &= \frac{1}{12} \end{aligned}$$

■

SUMMARY

A real-valued function defined on the outcome of a probability experiment is called a *random variable*.

If X is a random variable, then the function $F(x)$ defined by

$$F(x) = P\{X \leq x\}$$

is called the *distribution function* of X. All probabilities concerning X can be stated in terms of F.

A random variable whose set of possible values is either finite or countably infinite is called *discrete*. If X is a discrete random variable, then the function

$$p(x) = P\{X = x\}$$

is called the *probability mass function* of X. Also, the quantity $E[X]$ defined by

$$E[X] = \sum_{x:p(x)>0} xp(x)$$

is called the *expected value* of X. $E[X]$ is also commonly called the *mean* or the *expectation* of X.

A useful identity states that, for a function g,

$$E[g(X)] = \sum_{x:p(x)>0} g(x)p(x)$$

The *variance* of a random variable X, denoted by $\mathrm{Var}(X)$, is defined by

$$\mathrm{Var}(X) = E[(X - E[X])^2]$$

The variance, which is equal to the expected square of the difference between X and its expected value, is a measure of the spread of the possible values of X. A useful identity is

$$\mathrm{Var}(X) = E[X^2] - (E[X])^2$$

The quantity $\sqrt{\mathrm{Var}(X)}$ is called the *standard deviation* of X.

We now note some common types of discrete random variables. The random variable X whose probability mass function is given by

$$p(i) = \binom{n}{i} p^i(1 - p)^{n-i} \quad i = 0, \ldots, n$$

is said to be a binomial random variable with parameters n and p. Such a random variable can be interpreted as being the number of successes that occur when n independent trials, each of which results in a success with probability p, are performed. Its mean and variance are given by

$$E[X] = np \quad \mathrm{Var}(X) = np(1 - p)$$

The random variable X whose probability mass function is given by

$$p(i) = \frac{e^{-\lambda}\lambda^i}{i!} \quad i \geq 0$$

is said to be a *Poisson* random variable with parameter λ. If a large number of (approximately) independent trials are performed, each having a small probability of being successful, then the number of successful trials that result will have a distribution which is approximately that of a Poisson random variable. The mean and variance of a Poisson random variable are both equal to its parameter λ. That is,

$$E[X] = \mathrm{Var}(X) = \lambda$$

The random variable X whose probability mass function is given by

$$p(i) = p(1 - p)^{i-1} \quad i = 1, 2, \ldots$$

is said to be a *geometric* random variable with parameter p. Such a random variable represents the trial number of the first success when each trial is independently a success with probability p. Its mean and variance are given by

$$E[X] = \frac{1}{p} \quad \text{Var}(X) = \frac{1 - p}{p^2}$$

The random variable X whose probability mass function is given by

$$p(i) = \binom{i-1}{r-1} p^r (1-p)^{i-r} \quad i \geq r$$

is said to be a *negative binomial* random variable with parameters r and p. Such a random variable represents the trial number of the rth success when each trial is independently a success with probability p. Its mean and variance are given by

$$E[X] = \frac{r}{p} \quad \text{Var}(X) = \frac{r(1-p)}{p^2}$$

A *hypergeometric* random variable X with parameters n, N, and m represents the number of white balls selected when n balls are randomly chosen from an urn that contains N balls of which m are white. The probability mass function of this random variable is given by

$$p(i) = \frac{\binom{m}{i}\binom{N-m}{n-i}}{\binom{N}{n}} \quad i = 0, \ldots, m$$

With $p = m/N$, its mean and variance are

$$E[X] = np \quad \text{Var}(X) = \frac{N-n}{N-1} np(1-p)$$

An important property of the expected value is that the expected value of a sum of random variables is equal to the sum of their expected values. That is,

$$E\left[\sum_{i=1}^{n} X_i\right] = \sum_{i=1}^{n} E[X_i]$$

PROBLEMS

4.1. Two balls are chosen randomly from an urn containing 8 white, 4 black, and 2 orange balls. Suppose that we win \$2 for each black ball selected and we lose \$1 for each white ball selected. Let X denote our winnings. What are the possible values of X, and what are the probabilities associated with each value?

4.2. Two fair dice are rolled. Let X equal the product of the 2 dice. Compute $P\{X = i\}$ for $i = 1, \ldots, 36$.

4.3. Three dice are rolled. By assuming that each of the $6^3 = 216$ possible outcomes is equally likely, find the probabilities attached to the possible values that X can take on, where X is the sum of the 3 dice.

4.4. Five men and 5 women are ranked according to their scores on an examination. Assume that no two scores are alike and all 10! possible rankings are equally likely. Let X denote the highest ranking achieved by a woman. (For instance, $X = 1$

if the top-ranked person is female.) Find P{X = i}, $i = 1, 2, 3, \ldots, 8, 9, 10$.

4.5. Let X represent the difference between the number of heads and the number of tails obtained when a coin is tossed n times. What are the possible values of X?

4.6. In Problem 5, for $n = 3$, if the coin is assumed fair, what are the probabilities associated with the values that X can take on?

4.7. Suppose that a die is rolled twice. What are the possible values that the following random variables can take on:
- **(a)** the maximum value to appear in the two rolls;
- **(b)** the minimum value to appear in the two rolls;
- **(c)** the sum of the two rolls;
- **(d)** the value of the first roll minus the value of the second roll?

4.8. If the die in Problem 7 is assumed fair, calculate the probabilities associated with the random variables in parts (a) through (d).

4.9. Repeat Example 1b when the balls are selected with replacement.

4.10. In Example 1d, compute the conditional probability that we win i dollars, given that we win something; compute it for $i = 1, 2, 3$.

4.11. **(a)** An integer N is to be selected at random from $\{1, 2, \ldots, (10)^3\}$ in the sense that each integer has the same probability of being selected. What is the probability that N will be divisible by 3? by 5? by 7? by 15? by 105? How would your answer change if $(10)^3$ is replaced by $(10)^k$ as k became larger and larger?

(b) An important function in number theory—one whose properties can be shown to be related to what is probably the most important unsolved problem of mathematics, the Riemann hypothesis—is the Möbius function $\mu(n)$, defined for all positive integral values n as follows: Factor n into its prime factors. If there is a repeated prime factor, as in $12 = 2 \cdot 2 \cdot 3$ or $49 = 7 \cdot 7$, then $\mu(n)$ is defined to equal 0. Now let N be chosen at random from $\{1, 2, \ldots (10)^k\}$, where k is large. Determine $P\{\mu(N) = 0\}$ as $k \to \infty$.

Hint: To compute $P\{\mu(N) \neq 0\}$, use the identity

$$\prod_{i=1}^{\infty} \frac{P_i^2 - 1}{P_i^2} = \left(\frac{3}{4}\right)\left(\frac{8}{9}\right)\left(\frac{24}{25}\right)\left(\frac{48}{49}\right)\cdots = \frac{6}{\pi^2}$$

where P_i is the ith-smallest prime. (The number 1 is not a prime.)

4.12. In the game of Two-Finger Morra, 2 players show 1 or 2 fingers and simultaneously guess the number of fingers their opponent will show. If only one of the players guesses correctly, he wins an amount (in dollars) equal to the sum of the fingers shown by him and his opponent. If both players guess correctly or if neither guesses correctly, then no money is exchanged. Consider a specified player, and denote by X the amount of money he wins in a single game of Two-Finger Morra.
- **(a)** If each player acts independently of the other, and if each player makes his choice of the number of fingers he will hold up and the number he will guess that his opponent will hold up in such a way that each of the 4 possibilities is equally likely, what are the possible values of X and what are their associated probabilities?
- **(b)** Suppose that each player acts independently of the other. If each player decides to hold up the same number of fingers that he guesses his opponent will hold up, and if each player is equally likely to hold up 1 or 2 fingers, what are the possible values of X and their associated probabilities?

4.13. A salesman has scheduled two appointments to sell encyclopedias. His first appointment will lead to a sale with probability .3, and his second will lead independently to a sale with probability .6. Any sale made is equally likely to be either for the deluxe model, which costs \$1000, or the standard model, which costs \$500. Determine the probability mass function of X, the total dollar value of all sales.

4.14. Five distinct numbers are randomly distributed to players numbered 1 through 5. Whenever two players compare their numbers, the one with the higher one is declared the winner. Initially, players 1 and 2 compare their numbers; the winner then compares her number with that of player 3, and so on. Let X denote the number of times player 1 is a winner. Find $P\{X = i\}, i = 0, 1, 2, 3, 4$.

4.15. The National Basketball Association (NBA) draft lottery involves the 11 teams that had the worst won–lost records during the year. A total of 66 balls are placed in an urn. Each of these balls is inscribed with the name of a team: Eleven have the name of the team with the worst record, 10 have the name of the team with the second-worst record, 9 have the name of the team with the third-worst record, and so on (with 1 ball having the name of the team with the 11th-worst record). A ball is then chosen at random, and the team whose name is on the ball is given the first pick in the draft of players about to enter the league. Another ball is then chosen, and if it "belongs" to a team different from the one that received the first draft pick, then the team to which it belongs receives the second draft pick. (If the ball belongs

to the team receiving the first pick, then it is discarded and another one is chosen; this continues until the ball of another team is chosen.) Finally, another ball is chosen, and the team named on the ball (provided that it is different from the previous two teams) receives the third draft pick. The remaining draft picks 4 through 11 are then awarded to the 8 teams that did not "win the lottery," in inverse order of their won–lost records. For instance, if the team with the worst record did not receive any of the 3 lottery picks, then that team would receive the fourth draft pick. Let X denote the draft pick of the team with the worst record. Find the probability mass function of X.

4.16. In Problem 15, let team number 1 be the team with the worst record, let team number 2 be the team with the second-worst record, and so on. Let Y_i denote the team that gets draft pick number i. (Thus, $Y_1 = 3$ if the first ball chosen belongs to team number 3.) Find the probability mass function of (a) Y_1, (b) Y_2, and (c) Y_3.

4.17. Suppose that the distribution function of X is given by

$$F(b) = \begin{cases} 0 & b < 0 \\ \frac{b}{4} & 0 \le b < 1 \\ \frac{1}{2} + \frac{b-1}{4} & 1 \le b < 2 \\ \frac{11}{12} & 2 \le b < 3 \\ 1 & 3 \le b \end{cases}$$

(a) Find $P\{X = i\}, i = 1, 2, 3$.
(b) Find $P\{\frac{1}{2} < X < \frac{3}{2}\}$.

4.18. Four independent flips of a fair coin are made. Let X denote the number of heads obtained. Plot the probability mass function of the random variable $X - 2$.

4.19. If the distribution function of X is given by

$$F(b) = \begin{cases} 0 & b < 0 \\ \frac{1}{2} & 0 \le b < 1 \\ \frac{3}{5} & 1 \le b < 2 \\ \frac{4}{5} & 2 \le b < 3 \\ \frac{9}{10} & 3 \le b < 3.5 \\ 1 & b \ge 3.5 \end{cases}$$

calculate the probability mass function of X.

4.20. A gambling book recommends the following "winning strategy" for the game of roulette: Bet \$1 on red. If red appears (which has probability $\frac{18}{38}$), then take the \$1 profit and quit. If red does not appear and you lose this bet (which has probability $\frac{20}{38}$ of occurring), make additional \$1 bets on red on each of the next two spins of the roulette wheel and then quit. Let X denote your winnings when you quit.
(a) Find $P\{X > 0\}$.
(b) Are you convinced that the strategy is indeed a "winning" strategy? Explain your answer!
(c) Find $E[X]$.

4.21. Four buses carrying 148 students from the same school arrive at a football stadium. The buses carry, respectively, 40, 33, 25, and 50 students. One of the students is randomly selected. Let X denote the number of students that were on the bus carrying the randomly selected student. One of the 4 bus drivers is also randomly selected. Let Y denote the number of students on her bus.
(a) Which of $E[X]$ or $E[Y]$ do you think is larger? Why?
(b) Compute $E[X]$ and $E[Y]$.

4.22. Suppose that two teams play a series of games that ends when one of them has won i games. Suppose that each game played is, independently, won by team A with probability p. Find the expected number of games that are played when (a) $i = 2$ and (b) $i = 3$. Also, show in both cases that this number is maximized when $p = \frac{1}{2}$.

4.23. You have \$1000, and a certain commodity presently sells for \$2 per ounce. Suppose that after one week the commodity will sell for either \$1 or \$4 an ounce, with these two possibilities being equally likely.
(a) If your objective is to maximize the expected amount of money that you possess at the end of the week, what strategy should you employ?
(b) If your objective is to maximize the expected amount of the commodity that you possess at the end of the week, what strategy should you employ?

4.24. A and B play the following game: A writes down either number 1 or number 2, and B must guess which one. If the number that A has written down is i and B has guessed correctly, B receives i units from A. If B makes a wrong guess, B pays $\frac{3}{4}$ unit to A. If B randomizes his decision by guessing 1 with probability p and 2 with probability $1 - p$, determine his expected gain if (a) A has written down number 1 and (b) A has written down number 2.

What value of p maximizes the minimum possible value of B's expected gain, and what is this maximin value? (Note that B's expected

gain depends not only on p, but also on what A does.)

Consider now player A. Suppose that she also randomizes her decision, writing down number 1 with probability q. What is A's expected loss if (c) B chooses number 1 and (d) B chooses number 2?

What value of q minimizes A's maximum expected loss? Show that the minimum of A's maximum expected loss is equal to the maximum of B's minimum expected gain. This result, known as the minimax theorem, was first established in generality by the mathematician John von Neumann and is the fundamental result in the mathematical discipline known as the theory of games. The common value is called the value of the game to player B.

4.25. Two coins are to be flipped. The first coin will land on heads with probability .6, the second with probability .7. Assume that the results of the flips are independent, and let X equal the total number of heads that result.

(a) Find $P\{X = 1\}$.

(b) Determine $E[X]$.

4.26. One of the numbers 1 through 10 is randomly chosen. You are to try to guess the number chosen by asking questions with "yes–no" answers. Compute the expected number of questions you will need to ask in each of the following two cases:

(a) Your ith question is to be "Is it i?" $i = 1, 2, 3, 4, 5, 6, 7, 8, 9, 10$.

(b) With each question you try to eliminate one-half of the remaining numbers, as nearly as possible.

4.27. An insurance company writes a policy to the effect that an amount of money A must be paid if some event E occurs within a year. If the company estimates that E will occur within a year with probability p, what should it charge the customer in order that its expected profit will be 10 percent of A?

4.28. A sample of 3 items is selected at random from a box containing 20 items of which 4 are defective. Find the expected number of defective items in the sample.

4.29. There are two possible causes for a breakdown of a machine. To check the first possibility would cost C_1 dollars, and, if that were the cause of the breakdown, the trouble could be repaired at a cost of R_1 dollars. Similarly, there are costs C_2 and R_2 associated with the second possibility. Let p and $1 - p$ denote, respectively, the probabilities that the breakdown is caused by the first and second possibilities. Under what conditions on $p, C_i, R_i, i = 1, 2$, should we check the first possible cause of breakdown and then the second, as opposed to reversing the checking order, so as to minimize the expected cost involved in returning the machine to working order?

Note: If the first check is negative, we must still check the other possibility.

4.30. A person tosses a fair coin until a tail appears for the first time. If the tail appears on the nth flip, the person wins 2^n dollars. Let X denote the player's winnings. Show that $E[X] = +\infty$. This problem is known as the St. Petersburg paradox.

(a) Would you be willing to pay \$1 million to play this game once?

(b) Would you be willing to pay \$1 million for each game if you could play for as long as you liked and only had to settle up when you stopped playing?

4.31. Each night different meteorologists give us the probability that it will rain the next day. To judge how well these people predict, we will score each of them as follows: If a meteorologist says that it will rain with probability p, then he or she will receive a score of

$$\begin{array}{ll} 1 - (1 - p)^2 & \text{if it does rain} \\ 1 - p^2 & \text{if it does not rain} \end{array}$$

We will then keep track of scores over a certain time span and conclude that the meteorologist with the highest average score is the best predictor of weather. Suppose now that a given meteorologist is aware of our scoring mechanism and wants to maximize his or her expected score. If this person truly believes that it will rain tomorrow with probability p^*, what value of p should he or she assert so as to maximize the expected score?

4.32. To determine whether they have a certain disease, 100 people are to have their blood tested. However, rather than testing each individual separately, it has been decided first to place the people into groups of 10. The blood samples of the 10 people in each group will be pooled and analyzed together. If the test is negative, one test will suffice for the 10 people, whereas if the test is positive, each of the 10 people will also be individually tested and, in all, 11 tests will be made on this group. Assume that the probability that a person has the disease is .1 for all people, independently of each other, and compute the expected number of tests necessary for each group. (Note that we are assuming that the pooled test will be positive if at least one person in the pool has the disease.)

4.33. A newsboy purchases papers at 10 cents and sells them at 15 cents. However, he is not allowed to return unsold papers. If his daily demand is a binomial random variable with $n = 10, p = \frac{1}{3}$, approximately how many papers should he purchase so as to maximize his expected profit?

4.34. In Example 4b, suppose that the department store incurs an additional cost of c for each unit of unmet

demand. (This type of cost is often referred to as a goodwill cost because the store loses the goodwill of those customers whose demands it cannot meet.) Compute the expected profit when the store stocks s units, and determine the value of s that maximizes the expected profit.

4.35. A box contains 5 red and 5 blue marbles. Two marbles are withdrawn randomly. If they are the same color, then you win \$1.10; if they are different colors, then you win −\$1.00. (That is, you lose \$1.00.) Calculate
(a) the expected value of the amount you win;
(b) the variance of the amount you win.

4.36. Consider Problem 22 with $i = 2$. Find the variance of the number of games played, and show that this number is maximized when $p = \frac{1}{2}$.

4.37. Find $\text{Var}(X)$ and $\text{Var}(Y)$ for X and Y as given in Problem 21.

4.38. If $E[X] = 1$ and $\text{Var}(X) = 5$, find
(a) $E[(2 + X)^2]$;
(b) $\text{Var}(4 + 3X)$.

4.39. A ball is drawn from an urn containing 3 white and 3 black balls. After the ball is drawn, it is replaced and another ball is drawn. This process goes on indefinitely. What is the probability that, of the first 4 balls drawn, exactly 2 are white?

4.40. On a multiple-choice exam with 3 possible answers for each of the 5 questions, what is the probability that a student will get 4 or more correct answers just by guessing?

4.41. A man claims to have extrasensory perception. As a test, a fair coin is flipped 10 times and the man is asked to predict the outcome in advance. He gets 7 out of 10 correct. What is the probability that he would have done at least this well if he had no ESP?

4.42. Suppose that, in flight, airplane engines will fail with probability $1 - p$, independently from engine to engine. If an airplane needs a majority of its engines operative to complete a successful flight, for what values of p is a 5-engine plane preferable to a 3-engine plane?

4.43. A communications channel transmits the digits 0 and 1. However, due to static, the digit transmitted is incorrectly received with probability .2. Suppose that we want to transmit an important message consisting of one binary digit. To reduce the chance of error, we transmit 00000 instead of 0 and 11111 instead of 1. If the receiver of the message uses "majority" decoding, what is the probability that the message will be wrong when decoded? What independence assumptions are you making?

4.44. A satellite system consists of n components and functions on any given day if at least k of the n components function on that day. On a rainy day each of the components independently functions with probability p_1, whereas on a dry day they each independently function with probability p_2. If the probability of rain tomorrow is α, what is the probability that the satellite system will function?

4.45. A student is getting ready to take an important oral examination and is concerned about the possibility of having an "on" day or an "off" day. He figures that if he has an on day, then each of his examiners will pass him, independently of each other, with probability .8, whereas if he has an off day, this probability will be reduced to .4. Suppose that the student will pass the examination if a majority of the examiners pass him. If the student feels that he is twice as likely to have an off day as he is to have an on day, should he request an examination with 3 examiners or with 5 examiners?

4.46. Suppose that it takes at least 9 votes from a 12-member jury to convict a defendant. Suppose also that the probability that a juror votes a guilty person innocent is .2, whereas the probability that the juror votes an innocent person guilty is .1. If each juror acts independently and if 65 percent of the defendants are guilty, find the probability that the jury renders a correct decision. What percentage of defendants is convicted?

4.47. In some military courts, 9 judges are appointed. However, both the prosecution and the defense attorneys are entitled to a peremptory challenge of any judge, in which case that judge is removed from the case and is not replaced. A defendant is declared guilty if the majority of judges cast votes of guilty, and he or she is declared innocent otherwise. Suppose that when the defendant is, in fact, guilty, each judge will (independently) vote guilty with probability .7, whereas when the defendant is, in fact, innocent, this probability drops to .3.
(a) What is the probability that a guilty defendant is declared guilty when there are (i) 9, (ii) 8, and (iii) 7 judges?
(b) Repeat part (a) for an innocent defendant.
(c) If the prosecution attorney does not exercise the right to a peremptory challenge of a judge, and if the defense is limited to at most two such challenges, how many challenges should the defense attorney make if he or she is 60 percent certain that the client is guilty?

4.48. It is known that diskettes produced by a certain company will be defective with probability .01, independently of each other. The company sells the diskettes in packages of size 10 and offers a money-back guarantee that at most 1 of the 10 diskettes in the package will be defective. The guarantee is that the customer can return the entire package of diskettes if he or she finds more

than one defective diskette in it. If someone buys 3 packages, what is the probability that he or she will return exactly 1 of them?

4.49. When coin 1 is flipped, it lands on heads with probability .4; when coin 2 is flipped, it lands on heads with probability .7. One of these coins is randomly chosen and flipped 10 times.
(a) What is the probability that the coin lands on heads on exactly 7 of the 10 flips?
(b) Given that the first of these ten flips lands heads, what is the conditional probability that exactly 7 of the 10 flips land on heads?

4.50. Suppose that a biased coin that lands on heads with probability p is flipped 10 times. Given that a total of 6 heads results, find the conditional probability that the first 3 outcomes are
(a) h, t, t (meaning that the first flip results in heads, the second in tails, and the third in tails);
(b) t, h, t.

4.51. The expected number of typographical errors on a page of a certain magazine is .2. What is the probability that the next page you read contains (a) 0 and (b) 2 or more typographical errors? Explain your reasoning!

4.52. The monthly worldwide average number of airplane crashes of commercial airlines is 3.5. What is the probability that there will be
(a) at least 2 such accidents in the next month;
(b) at most 1 accident in the next month?
Explain your reasoning!

4.53. Approximately 80,000 marriages took place in the state of New York last year. Estimate the probability that, for at least one of these couples,
(a) both partners were born on April 30;
(b) both partners celebrated their birthday on the same day of the year.
State your assumptions.

4.54. Suppose that the average number of cars abandoned weekly on a certain highway is 2.2. Approximate the probability that there will be
(a) no abandoned cars in the next week;
(b) at least 2 abandoned cars in the next week.

4.55. A certain typing agency employs 2 typists. The average number of errors per article is 3 when typed by the first typist and 4.2 when typed by the second. If your article is equally likely to be typed by either typist, approximate the probability that it will have no errors.

4.56. How many people are needed so that the probability that at least one of them has the same birthday as you is greater than $\frac{1}{2}$?

4.57. Suppose that the number of accidents occurring on a highway each day is a Poisson random variable with parameter $\lambda = 3$.
(a) Find the probability that 3 or more accidents occur today.
(b) Repeat part (a) under the assumption that at least 1 accident occurs today.

4.58. Compare the Poisson approximation with the correct binomial probability for the following cases:
(a) $P\{X = 2\}$ when $n = 8, p = .1$;
(b) $P\{X = 9\}$ when $n = 10, p = .95$;
(c) $P\{X = 0\}$ when $n = 10, p = .1$;
(d) $P\{X = 4\}$ when $n = 9, p = .2$.

4.59. If you buy a lottery ticket in 50 lotteries, in each of which your chance of winning a prize is $\frac{1}{100}$, what is the (approximate) probability that you will win a prize
(a) at least once?
(b) exactly once?
(c) at least twice?

4.60. The number of times that a person contracts a cold in a given year is a Poisson random variable with parameter $\lambda = 5$. Suppose that a new wonder drug (based on large quantities of vitamin C) has just been marketed that reduces the Poisson parameter to $\lambda = 3$ for 75 percent of the population. For the other 25 percent of the population, the drug has no appreciable effect on colds. If an individual tries the drug for a year and has 2 colds in that time, how likely is it that the drug is beneficial for him or her?

4.61. The probability of being dealt a full house in a hand of poker is approximately .0014. Find an approximation for the probability that, in 1000 hands of poker, you will be dealt at least 2 full houses.

4.62. Consider n independent trials, each of which results in one of the outcomes $1, \ldots, k$ with respective probabilities $p_1, \ldots, p_k$, $\sum_{i=1}^{k} p_i = 1$. Show that if all the p_i are small, then the probability that no trial outcome occurs more than once is approximately equal to $\exp(-n(n-1)\sum_i p_i^2/2)$.

4.63. People enter a gambling casino at a rate of 1 every 2 minutes.
(a) What is the probability that no one enters between 12:00 and 12:05?
(b) What is the probability that at least 4 people enter the casino during that time?

4.64. The suicide rate in a certain state is 1 suicide per 100,000 inhabitants per month.
(a) Find the probability that, in a city of 400,000 inhabitants within this state, there will be 8 or more suicides in a given month.

(b) What is the probability that there will be at least 2 months during the year that will have 8 or more suicides?

(c) Counting the present month as month number 1, what is the probability that the first month to have 8 or more suicides will be month number $i, i \geq 1$?

What assumptions are you making?

4.65. Each of 500 soldiers in an army company independently has a certain disease with probability $1/10^3$. This disease will show up in a blood test, and to facilitate matters, blood samples from all 500 soldiers are pooled and tested.

(a) What is the (approximate) probability that the blood test will be positive (that is, at least one person has the disease)?

Suppose now that the blood test yields a positive result.

(b) What is the probability, under this circumstance, that more than one person has the disease?

One of the 500 people is Jones, who knows that he has the disease.

(c) What does Jones think is the probability that more than one person has the disease?

Because the pooled test was positive, the authorities have decided to test each individual separately. The first $i - 1$ of these tests were negative, and the ith one—which was on Jones—was positive.

(d) Given the preceding, scenario, what is the probability, as a function of i, that any of the remaining people have the disease?

4.66. A total of $2n$ people, consisting of n married couples, are randomly seated (all possible orderings being equally likely) at a round table. Let C_i denote the event that the members of couple i are seated next to each other, $i = 1, \ldots, n$.

(a) Find $P(C_i)$.

(b) For $j \neq i$, find $P(C_j|C_i)$.

(c) Approximate the probability, for n large, that there are no married couples who are seated next to each other.

4.67. Repeat the preceding problem when the seating is random but subject to the constraint that the men and women alternate.

4.68. In response to an attack of 10 missiles, 500 antiballistic missiles are launched. The missile targets of the antiballistic missiles are independent, and each antiballstic missile is equally likely to go towards any of the target missiles. If each antiballistic missile independently hits its target with probability .1, use the Poisson paradigm to approximate the probability that all missiles are hit.

4.69. A fair coin is flipped 10 times. Find the probability that there is a string of 4 consecutive heads by

(a) using the formula derived in the text;

(b) using the recursive equations derived in the text.

(c) Compare your answer with that given by the Poisson approximation.

4.70. At time 0, a coin that comes up heads with probability p is flipped and falls to the ground. Suppose it lands on heads. At times chosen according to a Poisson process with rate λ, the coin is picked up and flipped. (Between these times the coin remains on the ground.) What is the probability that the coin is on its head side at time t?

Hint What would be the conditional probability if there were no additional flips by time t, and what would it be if there were additional flips by time t?

4.71. Consider a roulette wheel consisting of 38 numbers 1 through 36, 0, and double 0. If Smith always bets that the outcome will be one of the numbers 1 through 12, what is the probability that

(a) Smith will lose his first 5 bets;

(b) his first win will occur on his fourth bet?

4.72. Two athletic teams play a series of games; the first team to win 4 games is declared the overall winner. Suppose that one of the teams is stronger than the other and wins each game with probability .6, independently of the outcomes of the other games. Find the probability, for $i = 4, 5, 6, 7$, that the stronger team wins the series in exactly i games. Compare the probability that the stronger team wins with the probability that it would win a 2-out-of-3 series.

4.73. Suppose in Problem 72 that the two teams are evenly matched and each has probability $\frac{1}{2}$ of winning each game. Find the expected number of games played.

4.74. An interviewer is given a list of people she can interview. If the interviewer needs to interview 5 people, and if each person (independently) agrees to be interviewed with probability $\frac{2}{3}$, what is the probability that her list of people will enable her to obtain her necessary number of interviews if the list consists of (a) 5 people and (b) 8 people? For part (b), what is the probability that the interviewer will speak to exactly (c) 6 people and (d) 7 people on the list?

4.75. A fair coin is continually flipped until heads appears for the 10th time. Let X denote the number of tails that occur. Compute the probability mass function of X.

4.76. Solve the Banach match problem (Example 8e) when the left-hand matchbox originally contained

N_1 matches and the right-hand box contained N_2 matches.

4.77. In the Banach matchbox problem, find the probability that, at the moment when the first box is emptied (as opposed to being found empty), the other box contains exactly k matches.

4.78. An urn contains 4 white and 4 black balls. We randomly choose 4 balls. If 2 of them are white and 2 are black, we stop. If not, we replace the balls in the urn and again randomly select 4 balls. This continues until exactly 2 of the 4 chosen are white. What is the probability that we shall make exactly n selections?

4.79. Suppose that a batch of 100 items contains 6 that are defective and 94 that are not defective. If X is the number of defective items in a randomly drawn sample of 10 items from the batch, find (a) $P\{X = 0\}$ and (b) $P\{X > 2\}$.

4.80. A game popular in Nevada gambling casinos is Keno, which is played as follows: Twenty numbers are selected at random by the casino from the set of numbers 1 through 80. A player can select from 1 to 15 numbers; a win occurs if some fraction of the player's chosen subset matches any of the 20 numbers drawn by the house. The payoff is a function of the number of elements in the player's selection and the number of matches. For instance, if the player selects only 1 number, then he or she wins if this number is among the set of 20, and the payoff is \$2.2 won for every dollar bet. (As the player's probability of winning in this case is $\frac{1}{4}$, it is clear that the "fair" payoff should be \$3 won for every \$1 bet.) When the player selects 2 numbers, a payoff (of odds) of \$12 won for every \$1 bet is made when both numbers are among the 20,

(a) What would be the fair payoff in this case? Let $P_{n,k}$ denote the probability that exactly k of the n numbers chosen by the player are among the 20 selected by the house.

(b) Compute $P_{n,k}$

(c) The most typical wager at Keno consists of selecting 10 numbers. For such a bet the casino pays off as shown in the following table. Compute the expected payoff:

Keno Payoffs in 10 Number Bets	
Number of matches	Dollars won for each \$1 bet
0–4	-1
5	1
6	17
7	179
8	1,299
9	2,599
10	24,999

4.81. In Example 8i, what percentage of i defective lots does the purchaser reject? Find it for $i = 1, 4$. Given that a lot is rejected, what is the conditional probability that it contained 4 defective components?

4.82. A purchaser of transistors buys them in lots of 20. It is his policy to randomly inspect 4 components from a lot and to accept the lot only if all 4 are nondefective. If each component in a lot is, independently, defective with probability .1, what proportion of lots is rejected?

4.83. There are three highways in the county. The number of daily accidents that occur on these highways are Poisson random variables with respective parameters .3, .5, and .7. Find the expected number of accidents that will happen on any of these highways today.

4.84. Suppose that 10 balls are put into 5 boxes, with each ball independently being put in box i with probability p_i, $\sum_{i=1}^{5} p_i = 1$.

(a) Find the expected number of boxes that do not have any balls.

(b) Find the expected number of boxes that have exactly 1 ball.

4.85. There are k types of coupons. Independently of the types of previously collected coupons, each new coupon collected is of type i with probability p_i, $\sum_{i=1}^{k} p_i = 1$. If n coupons are collected, find the expected number of distinct types that appear in this set. (That is, find the expected number of types of coupons that appear at least once in the set of n coupons.)

THEORETICAL EXERCISES

4.1. There are N distinct types of coupons, and each time one is obtained it will, independently of past choices, be of type i with probability $P_i, i = 1, \ldots, N$. Let T denote the number one need select to obtain at least one of each type. Compute $P\{T = n\}$.

Hint: Use an argument similar to the one used in Example 1e.

4.2. If X has distribution function F, what is the distribution function of e^X?

4.3. If X has distribution function F, what is the distribution function of the random variable $\alpha X + \beta$, where α and β are constants, $\alpha \neq 0$?

4.4. For a nonnegative integer-valued random variable N, show that

$$E[N] = \sum_{i=1}^{\infty} P\{N \geq i\}$$

Hint: $\sum_{i=1}^{\infty} P\{N \geq i\} = \sum_{i=1}^{\infty} \sum_{k=i}^{\infty} P\{N = k\}$. Now interchange the order of summation.

4.5. For a nonnegative integer-valued random variable N, show that

$$\sum_{i=0}^{\infty} iP\{N > i\} = \frac{1}{2}(E[N^2] - E[N])$$

Hint: $\sum_{i=0}^{\infty} iP\{N > i\} = \sum_{i=0}^{\infty} i \sum_{k=i+1}^{\infty} P\{N = k\}$. Now interchange the order of summation.

4.6. Let X be such that

$$P\{X = 1\} = p = 1 - P\{X = -1\}$$

Find $c \neq 1$ such that $E[c^X] = 1$.

4.7. Let X be a random variable having expected value μ and variance σ^2. Find the expected value and variance of

$$Y = \frac{X - \mu}{\sigma}$$

4.8. Find $\text{Var}(X)$ if

$$P(X = a) = p = 1 - P(X = b)$$

4.9. Show how the derivation of the binomial probabilities

$$P\{X = i\} = \binom{n}{i} p^i (1 - p)^{n-i}, \quad i = 0, \ldots, n$$

leads to a proof of the binomial theorem

$$(x + y)^n = \sum_{i=0}^{n} \binom{n}{i} x^i y^{n-i}$$

when x and y are nonnegative.
Hint: Let $p = \frac{x}{x+y}$.

4.10. Let X be a binomial random variable with parameters n and p. Show that

$$E\left[\frac{1}{X + 1}\right] = \frac{1 - (1 - p)^{n+1}}{(n + 1)p}$$

4.11. Consider n independent sequential trials, each of which is successful with probability p. If there is a total of k successes, show that each of the $n!/[k!(n - k)!]$ possible arrangements of the k successes and $n - k$ failures is equally likely.

4.12. There are n components lined up in a linear arrangement. Suppose that each component independently functions with probability p. What is the probability that no 2 neighboring components are both nonfunctional?
Hint: Condition on the number of defective components and use the results of Example 4c of Chapter 1.

4.13. Let X be a binomial random variable with parameters (n, p). What value of p maximizes $P\{X = k\}, k = 0, 1, \ldots, n$? This is an example of a statistical method used to estimate p when a binomial (n, p) random variable is observed to equal k. If we assume that n is known, then we estimate p by choosing that value of p which maximizes $P\{X = k\}$. This is known as the *method of maximum likelihood estimation*.

4.14. A family has n children with probability $\alpha p^n, n \geq 1$, where $\alpha \leq (1 - p)/p$.
(a) What proportion of families has no children?
(b) If each child is equally likely to be a boy or a girl (independently of each other), what proportion of families consists of k boys (and any number of girls)?

4.15. Suppose that n independent tosses of a coin having probability p of coming up heads are made. Show that the probability that an even number of heads results is $\frac{1}{2}[1 + (q - p)^n]$, where $q = 1 - p$. Do this by proving and then utilizing the identity

$$\sum_{i=0}^{[n/2]} \binom{n}{2i} p^{2i} q^{n-2i} = \frac{1}{2}\left[(p + q)^n + (q - p)^n\right]$$

where $[n/2]$ is the largest integer less than or equal to $n/2$. Compare this exercise with Theoretical Exercise 3.5 of Chapter 3.

4.16. Let X be a Poisson random variable with parameter λ. Show that $P\{X = i\}$ increases monotonically and then decreases monotonically as i increases, reaching its maximum when i is the largest integer not exceeding λ.
Hint: Consider $P\{X = i\}/P\{X = i - 1\}$.

4.17. Let X be a Poisson random variable with parameter λ.
(a) Show that

$$P\{X \text{ is even}\} = \frac{1}{2}\left[1 + e^{-2\lambda}\right]$$

by using the result of Theoretical Exercise 15 and the relationship between Poisson and binomial random variables.

(b) Verify the formula in part (a) directly by making use of the expansion of $e^{-\lambda} + e^{\lambda}$.

4.18. Let X be a Poisson random variable with parameter λ. What value of λ maximizes $P\{X = k\}, k \geq 0$?

4.19. Show that X is a Poisson random variable with parameter λ, then

$$E[X^n] = \lambda E[(X + 1)^{n-1}]$$

Now use this result to compute $E[X^3]$.

4.20. Consider n coins, each of which independently comes up heads with probability p. Suppose that n is large and p is small, and let $\lambda = np$. Suppose that all n coins are tossed; if at least one comes up heads, the experiment ends; if not, we again toss all n coins, and so on. That is, we stop the first time that at least one of the n coins come up heads. Let X denote the total number of heads that appear. Which of the following reasonings concerned with approximating $P\{X = 1\}$ is correct (in all cases, Y is a Poisson random variable with parameter λ)?

(a) Because the total number of heads that occur when all n coins are rolled is approximately a Poisson random variable with parameter λ,

$$P\{X = 1\} \approx P\{Y = 1\} = \lambda e^{-\lambda}$$

(b) Because the total number of heads that occur when all n coins are rolled is approximately a Poisson random variable with parameter λ, and because we stop only when this number is positive,

$$P\{X = 1\} \approx P\{Y = 1 | Y > 0\} = \frac{\lambda e^{-\lambda}}{1 - e^{-\lambda}}$$

(c) Because at least one coin comes up heads, X will equal 1 if none of the other $n - 1$ coins come up heads. Because the number of heads resulting from these $n - 1$ coins is approximately Poisson with mean $(n - 1)p \approx \lambda$,

$$P\{X = 1\} \approx P\{Y = 0\} = e^{-\lambda}$$

4.21. From a set of n randomly chosen people, let E_{ij} denote the event that persons i and j have the same birthday. Assume that each person is equally likely to have any of the 365 days of the year as his or her birthday. Find

(a) $P(E_{3,4}|E_{1,2})$;

(b) $P(E_{1,3}|E_{1,2})$;

(c) $P(E_{2,3}|E_{1,2} \cap E_{1,3})$.

What can you conclude from your answers to parts (a)–(c) about the independence of the $\binom{n}{2}$ events E_{ij}?

4.22. An urn contains $2n$ balls, of which 2 are numbered 1, 2 are numbered $2, \ldots,$ and 2 are numbered n. Balls are successively withdrawn 2 at a time without replacement. Let T denote the first selection in which the balls withdrawn have the same number (and let it equal infinity if none of the pairs withdrawn has the same number). We want to show that, for $0 < \alpha < 1$,

$$\lim_n P\{T > \alpha n\} = e^{-\alpha/2}$$

To verify the preceding formula, let M_k denote the number of pairs withdrawn in the first k selections, $k = 1, \ldots, n$.

(a) Argue that when n is large, M_k can be regarded as the number of successes in k (approximately) independent trials.

(b) Approximate $P\{M_k = 0\}$ when n is large.

(c) Write the event $\{T > \alpha n\}$ in terms of the value of one of the variables M_k.

(d) Verify the limiting probability given for $P\{T > \alpha n\}$.

4.23. Consider a random collection of n individuals. In approximating the probability that no 3 of these individuals share the same birthday, a better Poisson approximation than that obtained in the text (at least for values of n between 80 and 90) is obtained by letting E_i be the event that there are at least 3 birthdays on day i, $i = 1, \ldots, 365$.

(a) Find $P(E_i)$.

(b) Give an approximation for the probability that no 3 individuals share the same birthday.

(c) Evaluate the preceding when $n = 88$ (which can be shown to be the smallest value of n for which the probability exceeds .5).

4.24. Here is another way to obtain a set of recursive equations for determining P_n, the probability that there is a string of k consecutive heads in a sequence of n flips of a fair coin that comes up heads with probability p:

(a) Argue that, for $k < n$, there will be a string of k consecutive heads if either

1. there is a string of k consecutive heads within the first $n - 1$ flips, or
2. there is no string of k consecutive heads within the first $n - k - 1$ flips, flip $n - k$ is a tail, and flips $n - k + 1, \ldots, n$ are all heads.

(b) Using the preceding, relate P_n to P_{n-1}. Starting with $P_k = p^k$, the recursion can be used to obtain P_{k+1}, then P_{k+1}, and so on, up to P_n.

4.25. Suppose that the number of events that occur in a specified time is a Poisson random variable with parameter λ. If each event is counted with probability p, independently of every other event, show that the number of events that are counted is a Poisson random variable with parameter λp. Also, give an intuitive argument as to why this should be so. As an application of the preceding result, suppose that the number of distinct uranium deposits in a given area is a Poisson random variable with parameter $\lambda = 10$. If, in a fixed period of time, each deposit is discovered independently with probability $\frac{1}{50}$, find the probability that (a) exactly 1, (b) at least 1, and (c) at most 1 deposit is discovered during that time.

4.26. Prove

$$\sum_{i=0}^{n} e^{-\lambda}\frac{\lambda^i}{i!} = \frac{1}{n!}\int_{\lambda}^{\infty} e^{-x}x^n dx$$

Hint: Use integration by parts.

4.27. If X is a geometric random variable, show analytically that

$$P\{X = n + k | X > n\} = P\{X = k\}$$

Using the interpretation of a geometric random variable, give a verbal argument as to why the preceding equation is true.

4.28. Let X be a negative binomial random variable with parameters r and p, and let Y be a binomial random variable with parameters n and p. Show that

$$P\{X > n\} = P\{Y < r\}$$

Hint: Either one could attempt an analytical proof of the preceding equation, which is equivalent to proving the identity

$$\sum_{i=n+1}^{\infty}\binom{i-1}{r-1}p^r(1-p)^{i-r} = \sum_{i=0}^{r-1}\binom{n}{i} \times p^i(1-p)^{n-i}$$

or one could attempt a proof that uses the probabilistic interpretation of these random variables. That is, in the latter case, start by considering a sequence of independent trials having a common probability p of success. Then try to express the events $\{X > n\}$ and $\{Y < r\}$ in terms of the outcomes of this sequence.

4.29. For a hypergeometric random variable, determine

$$P\{X = k + 1\}/P\{X = k\}$$

4.30. Balls numbered 1 through N are in an urn. Suppose that $n, n \leq N$, of them are randomly selected without replacement. Let Y denote the largest number selected.

(a) Find the probability mass function of Y.

(b) Derive an expression for $E[Y]$ and then use Fermat's combinatorial identity (see Theoretical Exercise 11 of Chapter 1) to simplify the expression.

4.31. A jar contains $m + n$ chips, numbered $1, 2, \ldots, n + m$. A set of size n is drawn. If we let X denote the number of chips drawn having numbers that exceed each of the numbers of those remaining, compute the probability mass function of X.

4.32. A jar contains n chips. Suppose that a boy successively draws a chip from the jar, each time replacing the one drawn before drawing another. The process continues until the boy draws a chip that he has previously drawn. Let X denote the number of draws, and compute its probability mass function.

4.33. Show that Equation (8.6) follows from Equation (8.5).

4.34. From a set of n elements, a nonempty subset is chosen at random in the sense that all of the nonempty subsets are equally likely to be selected. Let X denote the number of elements in the chosen subset. Using the identities given in Theoretical Exercise 12 of Chapter 1, show that

$$E[X] = \frac{n}{2 - \left(\frac{1}{2}\right)^{n-1}}$$

$$\mathrm{Var}(X) = \frac{n \cdot 2^{2n-2} - n(n+1)2^{n-2}}{(2^n - 1)^2}$$

Show also that, for n large,

$$\mathrm{Var}(X) \backsim \frac{n}{4}$$

in the sense that the ratio $\mathrm{Var}(X)$ to $n/4$ approaches 1 as n approaches ∞. Compare this formula with the limiting form of $\mathrm{Var}(Y)$ when $P\{Y = i\} = 1/n, i = 1, \ldots, n$.

4.35. An urn initially contains one red and one blue ball. At each stage, a ball is randomly chosen and then replaced along with another of the same color. Let X denote the selection number of the first chosen ball that is blue. For instance, if the first selection is red and the second blue, then X is equal to 2.

(a) Find $P\{X > i\}, i \geq 1$.

(b) Show that, with probability 1, a blue ball is eventually chosen. (That is, show that $P\{X < \infty\} = 1$.)

(c) Find $E[X]$.

4.36. Suppose the possible values of X are $\{x_i\}$, the possible values of Y are $\{y_j\}$, and the possible values of $X + Y$ are $\{z_k\}$. Let A_k denote the set of all pairs of indices (i,j) such that $x_i + y_j = z_k$; that is, $A_k = \{(i,j) : x_i + y_j = z_k\}$.

(a) Argue that

$$P\{X + Y = z_k\} = \sum_{(i,j)\in A_k} P\{X = x_i, Y = y_j\}$$

(b) Show that

$$E[X + Y] = \sum_k \sum_{(i,j)\in A_k} (x_i + y_j)P\{X = x_i, Y = y_j\}$$

(c) Using the formula from part (b), argue that

$$E[X + Y] = \sum_i \sum_j (x_i + y_j)P\{X = x_i, Y = y_j\}$$

(d) Show that

$$P(X = x_i) = \sum_j P(X = x_i, Y = y_j),$$
$$P(Y = y_j) = \sum_i P\{X = x_i, Y = y_j\}$$

(e) Prove that

$$E[X + Y] = E[X] + E[Y]$$

SELF-TEST PROBLEMS AND EXERCISES

4.1. Suppose that the random variable X is equal to the number of hits obtained by a certain baseball player in his next 3 at bats. If $P\{X = 1\} = .3, P\{X = 2\} = .2$, and $P\{X = 0\} = 3P\{X = 3\}$, find $E[X]$.

4.2. Suppose that X takes on one of the values 0, 1, and 2. If for some constant $c, P\{X = i\} = cP\{X = i - 1\}, i = 1, 2$, find $E[X]$.

4.3. A coin that, when flipped, comes up heads with probability p is flipped until either heads or tails has occurred twice. Find the expected number of flips.

4.4. A certain community is composed of m families, n_i of which have i children, $\sum_{i=1}^{r} n_i = m$. If one of the families is randomly chosen, let X denote the number of children in that family. If one of the $\sum_{i=1}^{r} in_i$ children is randomly chosen, let Y denote the total number of children in the family of that child. Show that $E[Y] \geq E[X]$.

4.5. Suppose that $P\{X = 0\} = 1 - P\{X = 1\}$. If $E[X] = 3\text{Var}(X)$, find $P\{X = 0\}$.

4.6. There are 2 coins in a bin. When one of them is flipped, it lands on heads with probability .6, and when the other is flipped, it lands on heads with probability .3. One of these coins is to be randomly chosen and then flipped. Without knowing which coin is chosen, you can bet any amount up to 10 dollars, and you then either win that amount if the coin comes up heads or lose it if it comes up tails. Suppose, however, that an insider is willing to sell you, for an amount C, the information as to which coin was selected. What is your expected payoff if you buy this information? Note that if you buy it and then bet x, you will end up either winning $x - C$ or $-x - C$ (that is, losing $x + C$ in the latter case). Also, for what values of C does it pay to purchase the information?

4.7. A philanthropist writes a positive number x on a piece of red paper, shows the paper to an impartial observer, and then turns it face down on the table. The observer then flips a fair coin. If it shows heads, she writes the value $2x$ and, if tails, the value $x/2$, on a piece of blue paper, which she then turns face down on the table. Without knowing either the value x or the result of the coin flip, you have the option of turning over either the red or the blue piece of paper. After doing so and observing the number written on that paper, you may elect to receive as a reward either that amount or the (unknown) amount written on the other piece of paper. For instance, if you elect to turn over the blue paper and observe the value 100, then you can elect either to accept 100 as your reward or to take the amount (either 200 or 50) on the red paper. Suppose that you would like your expected reward to be large.

(a) Argue that there is no reason to turn over the red paper first, because if you do so, then no matter what value you observe, it is always better to switch to the blue paper.

(b) Let y be a fixed nonnegative value, and consider the following strategy: Turn over the blue paper, and if its value is at least y, then accept that amount. If it is less than y, then switch to the red paper. Let $R_y(x)$ denote the reward obtained if the philanthropist writes the amount x and you employ this strategy. Find $E[R_y(x)]$. Note that $E[R_0(x)]$ is the

expected reward if the philanthropist writes the amount x when you employ the strategy of always choosing the blue paper.

4.8. Let $B(n, p)$ represent a binomial random variable with parameters n and p. Argue that

$$P\{B(n,p) \le i\} = 1 - P\{B(n,1-p) \le n - i - 1\}$$

Hint: The number of successes less than or equal to i is equivalent to what statement about the number of failures?

4.9. If X is a binomial random variable with expected value 6 and variance 2.4, find $P\{X = 5\}$.

4.10. An urn contains n balls numbered 1 through n. If you withdraw m balls randomly in sequence, each time replacing the ball selected previously, find $P\{X = k\}, k = 1, \ldots, m$, where X is the maximum of the m chosen numbers.
Hint: First find $P\{X \le k\}$.

4.11. Teams A and B play a series of games, with the first team to win 3 games being declared the winner of the series. Suppose that team A independently wins each game with probability p. Find the conditional probability that team A wins
(a) the series given that it wins the first game;
(b) the first game given that it wins the series.

4.12. A local soccer team has 5 more games left to play. If it wins its game this weekend, then it will play its final 4 games in the upper bracket of its league, and if it loses, then it will play its final games in the lower bracket. If it plays in the upper bracket, then it will independently win each of its games in this bracket with probability .4, and if it plays in the lower bracket, then it will independently win each of its games with probability .7. If the probability that the team wins its game this weekend is .5, what is the probability that it wins at least 3 of its final 4 games?

4.13. Each of the members of a 7-judge panel independently makes a correct decision with probability .7. If the panel's decision is made by majority rule, what is the probability that the panel makes the correct decision? Given that 4 of the judges agreed, what is the probability that the panel made the correct decision?

4.14. On average, 5.2 hurricanes hit a certain region in a year. What is the probability that there will be 3 or fewer hurricanes hitting this year?

4.15. The number of eggs laid on a tree leaf by an insect of a certain type is a Poisson random variable with parameter λ. However, such a random variable can be observed only if it is positive, since if it is 0 then we cannot know that such an insect was on the leaf. If we let Y denote the observed number of eggs, then

$$P\{Y = i\} = P\{X = i | X > 0\}$$

where X is Poisson with parameter λ. Find $E[Y]$.

4.16. Each of n boys and n girls, independently and randomly, chooses a member of the other sex. If a boy and girl choose each other, they become a couple. Number the girls, and let G_i be the event that girl number i is part of a couple. Let $P_0 = 1 - P(\cup_{i=1}^{n} G_i)$ be the probability that no couples are formed.
(a) What is $P(G_i)$?
(b) What is $P(G_i|G_j)$?
(c) When n is large, approximate P_0.
(d) When n is large, approximate P_k, the probability that exactly k couples are formed.
(e) Use the inclusion–exclusion identity to evaluate P_0.

4.17. A total of $2n$ people, consisting of n married couples, are randomly divided into n pairs. Arbitrarily number the women, and let W_i denote the event that woman i is paired with her husband.
(a) Find $P(W_i)$.
(b) For $i \ne j$, find $P(W_i|W_j)$.
(c) When n is large, approximate the probability that no wife is paired with her husband.
(d) If each pairing must consist of a man and a woman, what does the problem reduce to?

4.18. A casino patron will continue to make \$5 bets on red in roulette until she has won 4 of these bets.
(a) What is the probability that she places a total of 9 bets?
(b) What is her expected winnings when she stops?
Remark: On each bet, she will either win \$5 with probability $\frac{18}{38}$ or lose \$5 with probability $\frac{20}{38}$.

4.19. When three friends go for coffee, they decide who will pay the check by each flipping a coin and then letting the "odd person" pay. If all three flips produce the same result (so that there is no odd person), then they make a second round of flips, and they continue to do so until there is an odd person. What is the probability that
(a) exactly 3 rounds of flips are made?
(b) more than 4 rounds are needed?

4.20. Show that if X is a geometric random variable with parameter p, then

$$E[1/X] = \frac{-p\log(p)}{1-p}$$

Hint: You will need to evaluate an expression of the form $\sum_{i=1}^{\infty} a^i/i$. To do so, write $a^i/i = \int_0^a x^{i-1}dx$, and then interchange the sum and the integral.

4.21. Suppose that

$$P\{X = a\} = p, \quad P\{X = b\} = 1 - p$$

(a) Show that $\frac{X-b}{a-b}$ is a Bernoulli random variable.
(b) Find Var(X).

4.22. Each game you play is a win with probability p. You plan to play 5 games, but if you win the fifth game, then you will keep on playing until you lose.
(a) Find the expected number of games that you play.
(b) Find the expected number of games that you lose.

4.23. Balls are randomly withdrawn, one at a time without replacement, from an urn that initially has N white and M black balls. Find the probability that n white balls are drawn before m black balls, $n \leq N, m \leq M$.

4.24. Ten balls are to be distributed among 5 urns, with each ball going into urn i with probability p_i, $\sum_{i=1}^{5} p_i = 1$. Let X_i denote the number of balls that go into urn i. Assume that events corresponding to the locations of different balls are independent.
(a) What type of random variable is X_i? Be as specific as possible.
(b) For $i \neq j$, what type of random variable is $X_i + X_j$?
(c) Find $P\{X_1 + X_2 + X_3 = 7\}$.

4.25. For the match problem (Example 5m in Chapter 2), find
(a) the expected number of matches.
(b) the variance of the number of matches.

4.26. Let α be the probability that a geometric random variable X with parameter p is an even number.
(a) Find α by using the identity $\alpha = \sum_{i=1}^{\infty} \mathrm{P\{X = 2i\}}$.
(b) Find α by conditioning on whether $X = 1$ or $X > 1$.

CHAPTER 5

Continuous Random Variables

5.1 INTRODUCTION
5.2 EXPECTATION AND VARIANCE OF CONTINUOUS RANDOM VARIABLES
5.3 THE UNIFORM RANDOM VARIABLE
5.4 NORMAL RANDOM VARIABLES
5.5 EXPONENTIAL RANDOM VARIABLES
5.6 OTHER CONTINUOUS DISTRIBUTIONS
5.7 THE DISTRIBUTION OF A FUNCTION OF A RANDOM VARIABLE

5.1 INTRODUCTION

In Chapter 4, we considered discrete random variables—that is, random variables whose set of possible values is either finite or countably infinite. However, there also exist random variables whose set of possible values is uncountable. Two examples are the time that a train arrives at a specified stop and the lifetime of a transistor. Let X be such a random variable. We say that X is a *continuous*[†] random variable if there exists a nonnegative function f, defined for all real $x \in (-\infty, \infty)$, having the property that, for any set B of real numbers,[‡]

$$P\{X \in B\} = \int_B f(x)\,dx \tag{1.1}$$

The function f is called the *probability density function* of the random variable X. (See Figure 5.1.)

In words, Equation (1.1) states that the probability that X will be in B may be obtained by integrating the probability density function over the set B. Since X must assume some value, f must satisfy

$$1 = P\{X \in (-\infty, \infty)\} = \int_{-\infty}^{\infty} f(x)\,dx$$

All probability statements about X can be answered in terms of f. For instance, from Equation (1.1), letting $B = [a, b]$, we obtain

$$P\{a \le X \le b\} = \int_a^b f(x)\,dx \tag{1.2}$$

[†]Sometimes called *absolutely continuous*.

[‡]Actually, for technical reasons Equation (1.1) is true only for the *measurable* sets B, which, fortunately, include all sets of practical interest.

CHAPTER 7

Properties of Expectation

7.1 INTRODUCTION
7.2 EXPECTATION OF SUMS OF RANDOM VARIABLES
7.3 MOMENTS OF THE NUMBER OF EVENTS THAT OCCUR
7.4 COVARIANCE, VARIANCE OF SUMS, AND CORRELATIONS
7.5 CONDITIONAL EXPECTATION
7.6 CONDITIONAL EXPECTATION AND PREDICTION
7.7 MOMENT GENERATING FUNCTIONS
7.8 ADDITIONAL PROPERTIES OF NORMAL RANDOM VARIABLES
7.9 GENERAL DEFINITION OF EXPECTATION

7.1 INTRODUCTION

In this chapter, we develop and exploit additional properties of expected values. To begin, recall that the expected value of the random variable X is defined by

$$E[X] = \sum_x xp(x)$$

where X is a discrete random variable with probability mass function $p(x)$, and by

$$E[X] = \int_{-\infty}^{\infty} xf(x)\,dx$$

when X is a continuous random variable with probability density function $f(x)$.

Since $E[X]$ is a weighted average of the possible values of X, it follows that if X must lie between a and b, then so must its expected value. That is, if

$$P\{a \le X \le b\} = 1$$

then

$$a \le E[X] \le b$$

To verify the preceding statement, suppose that X is a discrete random variable for which $P\{a \le X \le b\} = 1$. Since this implies that $p(x) = 0$ for all x outside of the interval $[a, b]$, it follows that

$$\begin{aligned} E[X] &= \sum_{x:p(x)>0} xp(x) \\ &\ge \sum_{x:p(x)>0} ap(x) \end{aligned}$$

$$= a \sum_{x:p(x)>0} p(x)$$
$$= a$$

In the same manner, it can be shown that $E[X] \leq b$, so the result follows for discrete random variables. As the proof in the continuous case is similar, the result follows.

7.2 EXPECTATION OF SUMS OF RANDOM VARIABLES

For a two-dimensional analog of Propositions 4.1 of Chapter 4 and 2.1 of Chapter 5, which give the computational formulas for the expected value of a function of a random variable, suppose that X and Y are random variables and g is a function of two variables. Then we have the following result.

Proposition 2.1. If X and Y have a joint probability mass function $p(x,y)$, then

$$E[g(X, Y)] = \sum_y \sum_x g(x, y)p(x, y)$$

If X and Y have a joint probability density function $f(x,y)$, then

$$E[g(X, Y)] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(x, y)f(x, y)\, dx\, dy$$

Let us give a proof of Proposition 2.1 when the random variables X and Y are jointly continuous with joint density function $f(x, y)$ and when $g(X, Y)$ is a nonnegative random variable. Because $g(X, Y) \geq 0$, we have, by Lemma 2.1 of Chapter 5, that

$$E[g(X, Y)] = \int_0^{\infty} P\{g(X, Y) > t\}\, dt$$

Writing

$$P\{g(X, Y) > t\} = \int \int_{(x,y):g(x,y)>t} f(x, y)\, dy\, dx$$

shows that

$$E[g(X, Y)] = \int_0^{\infty} \int \int_{(x,y):g(x,y)>t} f(x, y)\, dy\, dx\, dt$$

Interchanging the order of integration gives

$$E[g(X, Y) = \int_x \int_y \int_{t=0}^{g(x,y)} f(x, y)\, dt\, dy\, dx$$
$$= \int_x \int_y g(x, y)\, f(x, y)\, dy\, dx$$

Thus, the result is proven when $g(X, Y)$ is a nonnegative random variable. The general case then follows as in the one-dimensional case. (See Theoretical Exercises 2 and 3 of Chapter 5.)

EXAMPLE 2a

An accident occurs at a point X that is uniformly distributed on a road of length L. At the time of the accident, an ambulance is at a location Y that is also uniformly

distributed on the road. Assuming that X and Y are independent, find the expected distance between the ambulance and the point of the accident.

Solution. We need to compute $E[|X - Y|]$. Since the joint density function of X and Y is

$$f(x, y) = \frac{1}{L^2}, \quad 0 < x < L, \quad 0 < y < L$$

it follows from Proposition 2.1 that

$$E[|X - Y|] = \frac{1}{L^2} \int_0^L \int_0^L |x - y| \, dy \, dx$$

Now,

$$\begin{aligned} \int_0^L |x - y| dy &= \int_0^x (x - y) dy + \int_x^L (y - x) dy \\ &= \frac{x^2}{2} + \frac{L^2}{2} - \frac{x^2}{2} - x(L - x) \\ &= \frac{L^2}{2} + x^2 - xL \end{aligned}$$

Therefore,

$$\begin{aligned} E[|X - Y|] &= \frac{1}{L^2} \int_0^L \left(\frac{L^2}{2} + x^2 - xL \right) dx \\ &= \frac{L}{3} \end{aligned}$$

■

For an important application of Proposition 2.1, suppose that $E[X]$ and $E[Y]$ are both finite and let $g(X, Y) = X + Y$. Then, in the continuous case,

$$\begin{aligned} E[X + Y] &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (x + y) f(x, y) \, dx \, dy \\ &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x f(x, y) \, dy \, dx + \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} y f(x, y) \, dx \, dy \\ &= \int_{-\infty}^{\infty} x f_X(x) \, dx + \int_{-\infty}^{\infty} y f_Y(y) \, dy \\ &= E[X] + E[Y] \end{aligned}$$

The same result holds in general; thus, whenever $E[X]$ and $E[Y]$ are finite,

$$E[X + Y] = E[X] + E[Y] \tag{2.1}$$

EXAMPLE 2b

Suppose that, for random variables X and Y,

$$X \geq Y$$

That is, for any outcome of the probability experiment, the value of the random variable X is greater than or equal to the value of the random variable Y. Since $x \geq y$ is equivalent to the inequality $X - Y \geq 0$, it follows that $E[X - Y] \geq 0$, or, equivalently,

$$E[X] \geq E[Y]$$

■

Using Equation (2.1), we may show by a simple induction proof that if $E[X_i]$ is finite for all $i = 1, \ldots, n$, then

$$E[X_1 + \cdots + X_n] = E[X_1] + \cdots + E[X_n] \tag{2.2}$$

Equation (2.2) is an extremely useful formula whose utility will now be illustrated by a series of examples.

EXAMPLE 2c The sample mean

Let $X_1, \ldots, X_n$ be independent and identically distributed random variables having distribution function F and expected value μ. Such a sequence of random variables is said to constitute a sample from the distribution F. The quantity

$$\overline{X} = \sum_{i=1}^{n} \frac{X_i}{n}$$

is called the *sample mean*. Compute $E[\overline{X}]$.

Solution.

$$\begin{aligned} E[\overline{X}] &= E\left[\sum_{i=1}^{n} \frac{X_i}{n}\right] \\ &= \frac{1}{n} E\left[\sum_{i=1}^{n} X_i\right] \\ &= \frac{1}{n} \sum_{i=1}^{n} E[X_i] \\ &= \mu \quad \text{since } E[X_i] \equiv \mu \end{aligned}$$

That is, the expected value of the sample mean is μ, the mean of the distribution. When the distribution mean μ is unknown, the sample mean is often used in statistics to estimate it. ■

EXAMPLE 2d Boole's inequality

Let $A_1, \ldots, A_n$ denote events, and define the indicator variables $X_i, i = 1, \ldots, n$, by

$$X_i = \begin{cases} 1 & \text{if } A_i \text{ occurs} \\ 0 & \text{otherwise} \end{cases}$$

Let

$$X = \sum_{i=1}^{n} X_i$$

so X denotes the number of the events A_i that occur. Finally, let

$$Y = \begin{cases} 1 & \text{if } X \geq 1 \\ 0 & \text{otherwise} \end{cases}$$

so Y is equal to 1 if at least one of the A_i occurs and is 0 otherwise. Now, it is immediate that

$$X \geq Y$$

so

$$E[X] \geq E[Y]$$

But since

$$E[X] = \sum_{i=1}^{n} E[X_i] = \sum_{i=1}^{n} P(A_i)$$

and

$$E[Y] = P\{\text{at least one of the } A_i \text{ occur}\} = P\left(\bigcup_{i=1}^{n} A_i\right)$$

we obtain Boole's inequality, namely,

$$P\left(\bigcup_{i=1}^{n} A_i\right) \leq \sum_{i=1}^{n} P(A_i)$$ ■

The next three examples show how Equation (2.2) can be used to calculate the expected value of binomial, negative binomial, and hypergeometric random variables. These derivations should be compared with those presented in Chapter 4.

EXAMPLE 2e Expectation of a binomial random variable

Let X be a binomial random variable with parameters n and p. Recalling that such a random variable represents the number of successes in n independent trials when each trial has probability p of being a success, we have that

$$X = X_1 + X_2 + \cdots + X_n$$

where

$$X_i = \begin{cases} 1 & \text{if the } i\text{th trial is a success} \\ 0 & \text{if the } i\text{th trial is a failure} \end{cases}$$

Hence, X_i is a Bernoulli random variable having expectation $E[X_i] = 1(p) + 0(1 - p)$. Thus,

$$E[X] = E[X_1] + E[X_2] + \cdots + E[X_n] = np$$ ■

EXAMPLE 2f Mean of a negative binomial random variable

If independent trials having a constant probability p of being successes are performed, determine the expected number of trials required to amass a total of r successes.

Solution. If X denotes the number of trials needed to amass a total of r successes, then X is a negative binomial random variable that can be represented by

$$X = X_1 + X_2 + \cdots + X_r$$

where X_1 is the number of trials required to obtain the first success, X_2 the number of additional trials until the second success is obtained, X_3 the number of additional

trials until the third success is obtained, and so on. That is, X_i represents the number of additional trials required after the $(i-1)$st success until a total of i successes is amassed. A little thought reveals that each of the random variables X_i is a geometric random variable with parameter p. Hence, from the results of Example 8b of Chapter 4, $E[X_i] = 1/p, i = 1, 2, \ldots, r$; thus,

$$E[X] = E[X_1] + \cdots + E[X_r] = \frac{r}{p}$$ ■

EXAMPLE 2g Mean of a hypergeometric random variable

If n balls are randomly selected from an urn containing N balls of which m are white, find the expected number of white balls selected.

Solution. Let X denote the number of white balls selected, and represent X as

$$X = X_1 + \cdots + X_m$$

where

$$X_i = \begin{cases} 1 & \text{if the } i\text{th white ball is selected} \\ 0 & \text{otherwise} \end{cases}$$

Now

$$\begin{aligned} E[X_i] &= P\{X_i = 1\} \\ &= P\{i\text{th white ball is selected}\} \\ &= \frac{\binom{1}{1}\binom{N-1}{n-1}}{\binom{N}{n}} \\ &= \frac{n}{N} \end{aligned}$$

Hence,

$$E[X] = E[X_1] + \cdots + E[X_m] = \frac{mn}{N}$$

We could also have obtained the preceding result by using the alternative representation

$$X = Y_1 + \cdots + Y_n$$

where

$$Y_i = \begin{cases} 1 & \text{if the } i\text{th ball selected is white} \\ 0 & \text{otherwise} \end{cases}$$

Since the ith ball selected is equally likely to be any of the N balls, it follows that

$$E[Y_i] = \frac{m}{N}$$

so

$$E[X] = E[Y_1] + \cdots + E[Y_n] = \frac{nm}{N}$$ ■

EXAMPLE 2h Expected number of matches

Suppose that N people throw their hats into the center of a room. The hats are mixed up, and each person randomly selects one. Find the expected number of people that select their own hat.

Solution. Letting X denote the number of matches, we can compute $E[X]$ most easily by writing

$$X = X_1 + X_2 + \cdots + X_N$$

where

$$X_i = \begin{cases} 1 & \text{if the } i\text{th person selects his own hat} \\ 0 & \text{otherwise} \end{cases}$$

Since, for each i, the ith person is equally likely to select any of the N hats,

$$E[X_i] = P\{X_i = 1\} = \frac{1}{N}$$

Thus,

$$E[X] = E[X_1] + \cdots + E[X_N] = \left(\frac{1}{N}\right) N = 1$$

Hence, on the average, exactly one person selects his own hat. ■

EXAMPLE 2i Coupon-collecting problems

Suppose that there are N different types of coupons, and each time one obtains a coupon, it is equally likely to be any one of the N types. Find the expected number of coupons one need amass before obtaining a complete set of at least one of each type.

Solution. Let X denote the number of coupons collected before a complete set is attained. We compute $E[X]$ by using the same technique we used in computing the mean of a negative binomial random variable (Example 2f). That is, we define $X_i, i = 0, 1, \ldots, N - 1$ to be the number of additional coupons that need be obtained after i distinct types have been collected in order to obtain another distinct type, and we note that

$$X = X_0 + X_1 + \cdots + X_{N-1}$$

When i distinct types of coupons have already been collected, a new coupon obtained will be of a distinct type with probability $(N - i)/N$. Therefore,

$$P\{X_i = k\} = \frac{N - i}{N}\left(\frac{i}{N}\right)^{k-1} \qquad k \geq 1$$

or, in other words, X_i is a geometric random variable with parameter $(N - i)/N$.

Hence,

$$E[X_i] = \frac{N}{N - i}$$

implying that

$$\begin{aligned} E[X] &= 1 + \frac{N}{N-1} + \frac{N}{N-2} + \cdots + \frac{N}{1} \\ &= N\left[1 + \cdots + \frac{1}{N-1} + \frac{1}{N}\right] \end{aligned}$$

■

EXAMPLE 2j

Ten hunters are waiting for ducks to fly by. When a flock of ducks flies overhead, the hunters fire at the same time, but each chooses his target at random, independently of the others. If each hunter independently hits his target with probability p, compute the expected number of ducks that escape unhurt when a flock of size 10 flies overhead.

Solution. Let X_i equal 1 if the ith duck escapes unhurt and 0 otherwise, for $i = 1, 2, \ldots, 10$. The expected number of ducks to escape can be expressed as

$$E[X_1 + \cdots + X_{10}] = E[X_1] + \cdots + E[X_{10}]$$

To compute $E[X_i] = P\{X_i = 1\}$, we note that each of the hunters will, independently, hit the ith duck with probability $p/10$, so

$$P\{X_i = 1\} = \left(1 - \frac{p}{10}\right)^{10}$$

Hence,

$$E[X] = 10\left(1 - \frac{p}{10}\right)^{10}$$ ■

EXAMPLE 2k Expected number of runs

Suppose that a sequence of n 1's and m 0's is randomly permuted so that each of the $(n + m)!/(n!m!)$ possible arrangements is equally likely. Any consecutive string of 1's is said to constitute a run of 1's—for instance, if $n = 6, m = 4$, and the ordering is 1, 1, 1, 0, 1, 1, 0, 0, 1, 0, then there are 3 runs of 1's—and we are interested in computing the mean number of such runs. To compute this quantity, let

$$I_i = \begin{cases} 1 & \text{if a run of 1's starts at the } i\text{th position} \\ 0 & \text{otherwise} \end{cases}$$

Therefore, $R(1)$, the number of runs of 1, can be expressed as

$$R(1) = \sum_{i=1}^{n+m} I_i$$

and it follows that

$$E[R(1)] = \sum_{i=1}^{n+m} E[I_i]$$

Now,

$$\begin{aligned} E[I_1] &= P\{\text{“1” in position 1}\} \\ &= \frac{n}{n + m} \end{aligned}$$

and for $1 < i \leq n + m$,

$$\begin{aligned} E[I_i] &= P\{\text{“0” in position } i - 1, \text{“1” in position } i\} \\ &= \frac{m}{n+m}\frac{n}{n+m-1} \end{aligned}$$

Hence,

$$E[R(1)] = \frac{n}{n+m} + (n+m-1)\frac{nm}{(n+m)(n+m-1)}$$

Similarly, $E[R(0)]$, the expected number of runs of 0's, is

$$E[R(0)] = \frac{m}{n+m} + \frac{nm}{n+m}$$

and the expected number of runs of either type is

$$E[R(1) + R(0)] = 1 + \frac{2nm}{n+m}$$

■

EXAMPLE 2l A random walk in the plane

Consider a particle initially located at a given point in the plane, and suppose that it undergoes a sequence of steps of fixed length, but in a completely random direction. Specifically, suppose that the new position after each step is one unit of distance from the previous position and at an angle of orientation from the previous position that is uniformly distributed over $(0, 2\pi)$. (See Figure 7.1.) Compute the expected square of the distance from the origin after n steps.

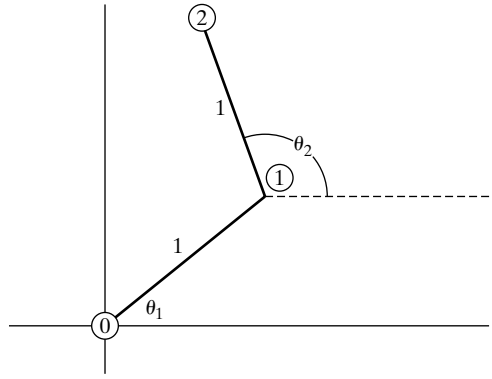


FIGURE 7.1

Solution. Letting (X_i, Y_i) denote the change in position at the ith step, $i = 1, \ldots, n$, in rectangular coordinates, we have

$$\begin{aligned} X_i &= \cos\theta_i \\ Y_i &= \sin\theta_i \end{aligned}$$

where $\theta_i, i = 1, \ldots, n$, are, by assumption, independent uniform $(0, 2\pi)$ random variables. Because the position after n steps has rectangular coordinates $\left(\sum_{i=1}^{n} X_i, \sum_{i=1}^{n} Y_i\right)$, it follows that D^2, the square of the distance from the origin, is given by

$$\begin{aligned} D^2 &= \left(\sum_{i=1}^{n} X_i\right)^2 + \left(\sum_{i=1}^{n} Y_i\right)^2 \\ &= \sum_{i=1}^{n} (X_i^2 + Y_i^2) + \sum\sum_{i \neq j} (X_i X_j + Y_i Y_j) \\ &= n + \sum\sum_{i \neq j} (\cos\theta_i \cos\theta_j + \sin\theta_i \sin\theta_j) \end{aligned}$$

where $\cos^2\theta_i + \sin^2\theta_i = 1$. Taking expectations and using the independence of θ_i and θ_j when $i \neq j$ and the fact that

$$\begin{aligned} 2\pi E[\cos\theta_i] &= \int_0^{2\pi} \cos u \, du = \sin 2\pi - \sin 0 = 0 \\ 2\pi E[\sin\theta_i] &= \int_0^{2\pi} \sin u \, du = \cos 0 - \cos 2\pi = 0 \end{aligned}$$

we arrive at

$$E[D^2] = n$$ ■

EXAMPLE 2m Analyzing the quick-sort algorithm

Suppose that we are presented with a set of n distinct values $x_1, x_2, \ldots, x_n$ and that we desire to put them in increasing order, or as it is commonly stated, to *sort* them. An efficient procedure for accomplishing this task is the quick-sort algorithm, which is defined as follows. When $n = 2$, the algorithm compares the two values and then puts them in the appropriate order. When $n > 2$, one of the elements is randomly chosen—say it is x_i—and then all of the other values are compared with x_i. Those smaller than x_i are put in a bracket to the left of x_i and those larger than x_i are put in a bracket to the right of x_i. The algorithm then repeats itself on these brackets and continues until all values have been sorted. For instance, suppose that we desire to sort the following 10 distinct values:

$$5, 9, 3, 10, 11, 14, 8, 4, 17, 6$$

We start by choosing one of them at random (that is, each value has probability $\frac{1}{10}$ of being chosen). Suppose, for instance, that the value 10 is chosen. We then compare each of the others to this value, putting in a bracket to the left of 10 all those values smaller than 10 and to the right all those larger. This gives

$$\{5, 9, 3, 8, 4, 6\}, 10, \{11, 14, 17\}$$

We now focus on a bracketed set that contains more than a single value—say the one on the left of the preceding—and randomly choose one of its values—say that 6 is chosen. Comparing each of the values in the bracket with 6 and putting the smaller

ones in a new bracket to the left of 6 and the larger ones in a bracket to the right of 6 gives

$$\{5,3,4\},6,\{9,8\},10,\{11,14,17\}$$

If we now consider the leftmost bracket, and randomly choose the value 4 for comparison then the next iteration yields

$$\{3\},4,\{5\},6,\{9,8\},10,\{11,14,17\}$$

This continues until there is no bracketed set that contains more than a single value.

If we let X denote the number of comparisons that it takes the quick-sort algorithm to sort n distinct numbers, then $E[X]$ is a measure of the effectiveness of this algorithm. To compute $E[X]$, we will first express X as a sum of other random variables as follows. To begin, give the following names to the values that are to be sorted: Let 1 stand for the smallest, let 2 stand for the next smallest, and so on. Then, for $1 \le i < j \le n$, let $I(i,j)$ equal 1 if i and j are ever directly compared, and let it equal 0 otherwise. With this definition, it follows that

$$X = \sum_{i=1}^{n-1} \sum_{j=i+1}^{n} I(i,j)$$

implying that

$$\begin{aligned} E[X] &= E\left[\sum_{i=1}^{n-1} \sum_{j=i+1}^{n} I(i,j)\right] \\ &= \sum_{i=1}^{n-1} \sum_{j=i+1}^{n} E[I(i,j)] \\ &= \sum_{i=1}^{n-1} \sum_{j=i+1}^{n} P\{i \text{ and } j \text{ are ever compared}\} \end{aligned}$$

To determine the probability that i and j are ever compared, note that the values $i, i+1, \ldots, j-1, j$ will initially be in the same bracket (since all values are initially in the same bracket) and will remain in the same bracket if the number chosen for the first comparison is not between i and j. For instance, if the comparison number is larger than j, then all the values $i, i+1, \ldots, j-1, j$ will go in a bracket to the left of the comparison number, and if it is smaller than i, then they will all go in a bracket to the right. Thus all the values $i, i+1, \ldots, j-1, j$ will remain in the same bracket until the first time that one of them is chosen as a comparison value. At that point all the other values between i and j will be compared with this comparison value. Now, if this comparison value is neither i nor j, then upon comparison with it, i will go into a left bracket and j into a right bracket, and thus i and j will be in different brackets and so will never be compared. On the other hand, if the comparison value of the set $i, i+1, \ldots, j-1, j$ is either i or j, then there will be a direct comparison between i and j. Now, given that the comparison value is one of the values between i and j, it follows that it is equally likely to be any of these $j - i + 1$ values, and thus the probability that it is either i or j is $2/(j - i + 1)$. Therefore, we can conclude that

$$P\{i \text{ and } j \text{ are ever compared}\} = \frac{2}{j - i + 1}$$

and

$$E[X] = \sum_{i=1}^{n-1} \sum_{j=i+1}^{n} \frac{2}{j - i + 1}$$

To obtain a rough approximation of the magnitude of $E[X]$ when n is large, we can approximate the sums by integrals. Now

$$\begin{aligned}\sum_{j=i+1}^{n} \frac{2}{j - i + 1} &\approx \int_{i+1}^{n} \frac{2}{x - i + 1}\, dx \\ &= 2\log(x - i + 1)\Big|_{i+1}^{n} \\ &= 2\log(n - i + 1) - 2\log(2) \\ &\approx 2\log(n - i + 1)\end{aligned}$$

Thus

$$\begin{aligned}E[X] &\approx \sum_{i=1}^{n-1} 2\log(n - i + 1) \\ &\approx 2\int_{1}^{n-1} \log(n - x + 1)\, dx \\ &= 2\int_{2}^{n} \log(y)dy \\ &= 2(y\log(y) - y)\Big|_{2}^{n} \\ &\approx 2n\log(n)\end{aligned}$$

Thus we see that when n is large, the quick-sort algorithm requires, on average, approximately $2n\log(n)$ comparisons to sort n distinct values. ■

EXAMPLE 2n The probability of a union of events

Let $A_1, \ldots A_n$ denote events, and define the indicator variables $X_i, i = 1, \ldots, n$, by

$$X_i = \begin{cases} 1 & \text{if } A_i \text{ occurs} \\ 0 & \text{otherwise} \end{cases}$$

Now, note that

$$1 - \prod_{i=1}^{n}(1 - X_i) = \begin{cases} 1 & \text{if } \cup A_i \text{ occurs} \\ 0 & \text{otherwise} \end{cases}$$

Hence,

$$E\left[1 - \prod_{i=1}^{n}(1 - X_i)\right] = P\left(\bigcup_{i=1}^{n} A_i\right)$$

Expanding the left side of the preceding formula yields

$$P\left(\bigcup_{i=1}^{n} A_i\right) = E\left[\sum_{i=1}^{n} X_i - \sum\sum_{i<j} X_i X_j + \sum\sum\sum_{i<j<k} X_i X_j X_k - \cdots + (-1)^{n+1} X_1 \cdots X_n\right] \tag{2.3}$$

However,

$$X_{i_1} X_{i_2} \cdots X_{i_k} = \begin{cases} 1 & \text{if } A_{i_1} A_{i_2} \cdots A_{i_k} \text{ occurs} \\ 0 & \text{otherwise} \end{cases}$$

so

$$E[X_{i_1} \cdots X_{i_k}] = P(A_{i_1} \cdots A_{i_k})$$

Thus, Equation (2.3) is just a statement of the well-known formula for the union of events:

$$P(\cup A_i) = \sum_i P(A_i) - \sum\sum_{i<j} P(A_i A_j) + \sum\sum\sum_{i<j<k} P(A_i A_j A_k) - \cdots + (-1)^{n+1} P(A_1 \cdots A_n)$$

■

When one is dealing with an infinite collection of random variables $X_i, i \geq 1$, each having a finite expectation, it is not necessarily true that

$$E\left[\sum_{i=1}^{\infty} X_i\right] = \sum_{i=1}^{\infty} E[X_i] \tag{2.4}$$

To determine when (2.4) is valid, we note that $\sum_{i=1}^{\infty} X_i = \lim_{n\to\infty} \sum_{i=1}^{n} X_i$. Thus,

$$\begin{aligned} E\left[\sum_{i=1}^{\infty} X_i\right] &= E\left[\lim_{n\to\infty} \sum_{i=1}^{n} X_i\right] \\ &\overset{?}{=} \lim_{n\to\infty} E\left[\sum_{i=1}^{n} X_i\right] \\ &= \lim_{n\to\infty} \sum_{i=1}^{n} E[X_i] \\ &= \sum_{i=1}^{\infty} E[X_i] \end{aligned} \tag{2.5}$$

Hence, Equation (2.4) is valid whenever we are justified in interchanging the expectation and limit operations in Equation (2.5). Although, in general, this interchange is *not* justified, it can be shown to be valid in two important special cases:

1. The X_i are all nonnegative random variables. (That is, $P\{X_i \geq 0\} = 1$ for all i.)
2. $\sum_{i=1}^{\infty} E[|X_i|] < \infty$.

EXAMPLE 2o

Consider any nonnegative, integer-valued random variable X. If, for each $i \geq 1$, we define

$$X_i = \begin{cases} 1 & \text{if } X \geq i \\ 0 & \text{if } X < i \end{cases}$$

then

$$\begin{aligned} \sum_{i=1}^{\infty} X_i &= \sum_{i=1}^{X} X_i + \sum_{i=X+1}^{\infty} X_i \\ &= \sum_{i=1}^{X} 1 + \sum_{i=X+1}^{\infty} 0 \\ &= X \end{aligned}$$

Hence, since the X_i are all nonnegative, we obtain

$$\begin{aligned} E[X] &= \sum_{i=1}^{\infty} E(X_i) \\ &= \sum_{i=1}^{\infty} P\{X \geq i\} \end{aligned} \tag{2.6}$$

a useful identity. ■

EXAMPLE 2p

Suppose that n elements—call them $1, 2, \ldots, n$—must be stored in a computer in the form of an ordered list. Each unit of time, a request will be made for one of these elements—i being requested, independently of the past, with probability $P(i)$, $i \geq 1$, $\sum_i P(i) = 1$. Assuming that these probabilities are known, what ordering minimizes the average position in the line of the element requested?

Solution. Suppose that the elements are numbered so that $P(1) \geq P(2) \geq \cdots \geq P(n)$. To show that $1, 2, \ldots, n$ is the optimal ordering, let X denote the position of the requested element. Now, under any ordering—say, $O = i_1, i_2, \ldots, i_n$,

$$\begin{aligned} P_O\{X \geq k\} &= \sum_{j=k}^{n} P(i_j) \\ &\geq \sum_{j=k}^{n} P(j) \\ &= P_{1,2,\ldots,n}\{X \geq k\} \end{aligned}$$

Summing over k and using Equation (2.6) yields

$$E_o[X] \geq E_{1,2,\ldots,n}[X]$$

thus showing that ordering the elements in decreasing order of the probability that they are requested minimizes the expected position of the element requested. ■

*7.2.1 Obtaining Bounds from Expectations via the Probabilistic Method

The probabilistic method is a technique for analyzing the properties of the elements of a set by introducing probabilities on the set and then studying an element chosen according to those probabilities. The technique was previously seen in Example 4l of Chapter 3, where it was used to show that a set contained an element that satisfied a certain property. In this subsection, we show how it can sometimes be used to bound complicated functions.

Let f be a function on the elements of a finite set $\mathcal{S}$, and suppose that we are interested in

$$m = \max_{s \in \mathcal{S}} f(s)$$

A useful lower bound for m can often be obtained by letting S be a random element of $\mathcal{S}$ for which the expected value of $f(S)$ is computable and then noting that $m \geq f(S)$ implies that

$$m \geq E[f(S)]$$

with strict inequality if $f(S)$ is not a constant random variable. That is, $E[f(S)]$ is a lower bound on the maximum value.

EXAMPLE 2q The maximum number of Hamiltonian paths in a tournament

A round-robin tournament of $n > 2$ contestants is a tournament in which each of the $\binom{n}{2}$ pair of contestants play each other exactly once. Suppose that the players are numbered $1, 2, 3, \ldots, n$. The permutation $i_1, i_2, \ldots i_n$ is said to be a *Hamiltonian path* if i_1 beats i_2, i_2 beats i_3, ..., and i_{n-1} beats i_n. A problem of some interest is to determine the largest possible number of Hamiltonian paths.

As an illustration, suppose that there are 3 players. On the one hand, one of them wins twice, then there is a single Hamiltonian path. (For instance, if 1 wins twice and 2 beats 3, then the only Hamiltonian path is 1, 2, 3.) On the other hand, if each of the players wins once, than there are 3 Hamiltonian paths. (For instance, if 1 beats 2, 2 beats 3, and 3 beats 1, then 1, 2, 3; 2, 3, 1; and 3, 1, 2, are all Hamiltonians). Hence, when $n = 3$, there is a maximum of 3 Hamiltonian paths.

We now show that there is an outcome of the tournament that results in more than $n!/2^{n-1}$ Hamiltonian paths. To begin, let the outcome of the tournament specify the result of each of the $\binom{n}{2}$ games played, and let $\mathcal{S}$ denote the set of all $2^{\binom{n}{2}}$ possible tournament outcomes. Then, with $f(s)$ defined as the number of Hamiltonian paths that result when the outcome is $s \in \mathcal{S}$, we are asked to show that

$$\max_s f(s) \geq \frac{n!}{2^{n-1}}$$

To show this, consider the randomly chosen outcome S that is obtained when the results of the $\binom{n}{2}$ games are independent, with each contestant being equally likely to win each encounter. To determine $E[f(S)]$, the expected number of Hamiltonian paths that result from the outcome S, number the $n!$ permutations, and, for $i = 1, \ldots, n!$, let

$$X_i = \begin{cases} 1, & \text{if permutation } i \text{ is a Hamiltonian} \\ 0, & \text{otherwise} \end{cases}$$

Now, conditional on the event $S_{i,j}$, both A_i and A_j will occur if, at the time the first type T_j coupon is collected, of the $n - j + 2$ types consisting of types T_i, T_j, and the $n - j$ as yet uncollected types, T_i and T_j are both collected after the other $n - j$. But this implies that

$$P(A_iA_j|S_{i,j}) = 2\,\frac{1}{n - j + 2}\,\frac{1}{n - j + 1}$$

Therefore,

$$P(A_iA_j) = \frac{2}{(n + 1 - i)(n + 2 - j)}, \quad i < j$$

yielding

$$E[X(X - 1)] = 4\sum_{i<j} \frac{1}{(n + 1 - i)(n + 2 - j)}$$

Consequently, using the previous result for $E[X]$, we obtain

$$\mathrm{Var}(X) = 4\sum_{i<j} \frac{1}{(n + 1 - i)(n + 2 - j)} + \sum_{i=1}^{n} \frac{1}{i} - \left(\sum_{i=1}^{n} \frac{1}{i}\right)^2 \qquad \blacksquare$$

7.4 COVARIANCE, VARIANCE OF SUMS, AND CORRELATIONS

The following proposition shows that the expectation of a product of independent random variables is equal to the product of their expectations.

Proposition 4.1. If X and Y are independent, then, for any functions h and g,

$$E[g(X)h(Y)] = E[g(X)]E[h(Y)]$$

Proof. Suppose that X and Y are jointly continuous with joint density $f(x, y)$. Then

$$\begin{aligned} E[g(X)h(Y)] &= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} g(x)h(y)f(x, y)\,dx\,dy \\ &= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} g(x)h(y)f_X(x)f_Y(y)\,dx\,dy \\ &= \int_{-\infty}^{\infty} h(y)f_Y(y)dy \int_{-\infty}^{\infty} g(x)f_X(x)\,dx \\ &= E[h(Y)]E[g(X)] \end{aligned}$$

The proof in the discrete case is similar. □

Just as the expected value and the variance of a single random variable give us information about that random variable, so does the covariance between two random variables give us information about the relationship between the random variables.

Definition

The covariance between X and Y, denoted by Cov (X, Y), is defined by

$$\mathrm{Cov}(X, Y) = E[(X - E[X])(Y - E[Y])]$$

Upon expanding the right side of the preceding definition, we see that

$$\begin{aligned}\text{Cov}(X, Y) &= E[XY - E[X]Y - XE[Y] + E[Y]E[X]] \\ &= E[XY] - E[X]E[Y] - E[X]E[Y] + E[X]E[Y] \\ &= E[XY] - E[X]E[Y]\end{aligned}$$

Note that if X and Y are independent, then, by Proposition 4.1, $\text{Cov}(X, Y) = 0$. However, the converse is not true. A simple example of two dependent random variables X and Y having zero covariance is obtained by letting X be a random variable such that

$$P\{X = 0\} = P\{X = 1\} = P\{X = -1\} = \frac{1}{3}$$

and defining

$$Y = \begin{cases} 0 & \text{if } X \neq 0 \\ 1 & \text{if } X = 0 \end{cases}$$

Now, $XY = 0$, so $E[XY] = 0$. Also, $E[X] = 0$. Thus,

$$\text{Cov}(X, Y) = E[XY] - E[X]E[Y] = 0$$

However, X and Y are clearly not independent.

The following proposition lists some of the properties of covariance.

Proposition 4.2.

(i) $\text{Cov}(X, Y) = \text{Cov}(Y, X)$

(ii) $\text{Cov}(X, X) = \text{Var(X)}$

(iii) $\text{Cov}(aX, Y) = a\,\text{Cov}(X, Y)$

(iv) $$\text{Cov}\left(\sum_{i=1}^{n} X_i, \sum_{j=1}^{m} Y_j\right) = \sum_{i=1}^{n}\sum_{j=1}^{m} \text{Cov}(X_i, Y_j)$$

Proof of Proposition 4.2: Parts (i) and (ii) follow immediately from the definition of covariance, and part (iii) is left as an exercise for the reader. To prove part (iv), which states that the covariance operation is additive (as is the operation of taking expectations), let $\mu_i = E[X_i]$ and $v_j = E[Y_j]$. Then

$$E\left[\sum_{i=1}^{n} X_i\right] = \sum_{i=1}^{n} \mu_i, \quad E\left[\sum_{j=1}^{m} Y_j\right] = \sum_{j=1}^{m} v_j$$

and

$$\begin{aligned}\text{Cov}\left(\sum_{i=1}^{n} X_i, \sum_{j=1}^{m} Y_j\right) &= E\left[\left(\sum_{i=1}^{n} X_i - \sum_{i=1}^{n} \mu_i\right)\left(\sum_{j=1}^{m} Y_j - \sum_{j=1}^{m} v_j\right)\right] \\ &= E\left[\sum_{i=1}^{n}(X_i - \mu_i)\sum_{j=1}^{m}(Y_j - v_j)\right]\end{aligned}$$

$$= E\left[\sum_{i=1}^{n}\sum_{j=1}^{m}(X_i - \mu_i)(Y_j - v_j)\right]$$
$$= \sum_{i=1}^{n}\sum_{j=1}^{m}E[(X_i - \mu_i)(Y_j - v_j)]$$

where the last equality follows because the expected value of a sum of random variables is equal to the sum of the expected values. □

It follows from parts (ii) and (iv) of Proposition 4.2, upon taking $Y_j = X_j, j = 1, \ldots, n$, that

$$\text{Var}\left(\sum_{i=1}^{n}X_i\right) = \text{Cov}\left(\sum_{i=1}^{n}X_i, \sum_{j=1}^{n}X_j\right)$$
$$= \sum_{i=1}^{n}\sum_{j=1}^{n}\text{Cov}(X_i, X_j)$$
$$= \sum_{i=1}^{n}\text{Var}(X_i) + \sum\sum_{i\neq j}\text{Cov}(X_i, X_j)$$

Since each pair of indices $i, j, i \neq j$, appears twice in the double summation, the preceding formula is equivalent to

$$\boxed{\text{Var}\left(\sum_{i=1}^{n}X_i\right) = \sum_{i=1}^{n}\text{Var}(X_i) + 2\sum\sum_{i<j}\text{Cov}(X_i, X_j)} \tag{4.1}$$

If $X_1, \ldots, X_n$ are pairwise independent, in that X_i and X_j are independent for $i \neq j$, then Equation (4.1) reduces to

$$\text{Var}\left(\sum_{i=1}^{n}X_i\right) = \sum_{i=1}^{n}\text{Var}(X_i)$$

The following examples illustrate the use of Equation (4.1).

EXAMPLE 4a

Let $X_1, \ldots, X_n$ be independent and identically distributed random variables having expected value μ and variance σ^2, and as in Example 2c, let $\overline{X} = \sum_{i=1}^{n} X_i/n$ be the sample mean. The quantities $X_i - \overline{X}, i = 1, \ldots, n$, are called *deviations*, as they equal the differences between the individual data and the sample mean. The random variable

$$S^2 = \sum_{i=1}^{n}\frac{(X_i - \overline{X})^2}{n - 1}$$

is called the *sample variance*. Find (a) $\text{Var}(\overline{X})$ and (b) $E[S^2]$.

Solution.

$$\begin{aligned}
\text{(a)}\quad \text{Var}(\overline{X}) &= \left(\frac{1}{n}\right)^2 \text{Var}\left(\sum_{i=1}^{n} X_i\right) \\
&= \left(\frac{1}{n}\right)^2 \sum_{i=1}^{n} \text{Var}(X_i) \quad \text{by independence} \\
&= \frac{\sigma^2}{n}
\end{aligned}$$

(b) We start with the following algebraic identity:

$$\begin{aligned}
(n-1)S^2 &= \sum_{i=1}^{n} (X_i - \mu + \mu - \overline{X})^2 \\
&= \sum_{i=1}^{n} (X_i - \mu)^2 + \sum_{i=1}^{n} (\overline{X} - \mu)^2 - 2(\overline{X} - \mu) \sum_{i=1}^{n} (X_i - \mu) \\
&= \sum_{i=1}^{n} (X_i - \mu)^2 + n(\overline{X} - \mu)^2 - 2(\overline{X} - \mu) n(\overline{X} - \mu) \\
&= \sum_{i=1}^{n} (X_i - \mu)^2 - n(\overline{X} - \mu)^2
\end{aligned}$$

Taking expectations of the preceding yields

$$\begin{aligned}
(n-1)E[S^2] &= \sum_{i=1}^{n} E[(X_i - \mu)^2] - nE[(\overline{X} - \mu)^2] \\
&= n\sigma^2 - n\text{Var}(\overline{X}) \\
&= (n-1)\sigma^2
\end{aligned}$$

where the final equality made use of part (a) of this example and the one preceding it made use of the result of Example 2c, namely, that $E[\overline{X}] = \mu$. Dividing through by $n - 1$ shows that the expected value of the sample variance is the distribution variance σ^2. ■

Our next example presents another method for obtaining the variance of a binomial random variable.

EXAMPLE 4b Variance of a binomial random variable

Compute the variance of a binomial random variable X with parameters n and p.

Solution. Since such a random variable represents the number of successes in n independent trials when each trial has the common probability p of being a success, we may write

$$X = X_1 + \cdots + X_n$$

where the X_i are independent Bernoulli random variables such that

$$X_i = \begin{cases} 1 & \text{if the } i\text{th trial is a success} \\ 0 & \text{otherwise} \end{cases}$$

Hence, from Equation (4.1), we obtain

$$\text{Var}(X) = \text{Var}(X_1) + \cdots + \text{Var}(X_n)$$

But

$$\begin{aligned}\text{Var}(X_i) &= E[X_i^2] - (E[X_i])^2 \\ &= E[X_i] - (E[X_i])^2 \quad \text{since } X_i^2 = X_i \\ &= p - p^2\end{aligned}$$

Thus,

$$\text{Var}(X) = np(1 - p)$$ ■

EXAMPLE 4c Sampling from a finite population

Consider a set of N people, each of whom has an opinion about a certain subject that is measured by a real number v that represents the person's "strength of feeling" about the subject. Let v_i represent the strength of feeling of person i, $i = 1, \ldots N$.

Suppose that the quantities $v_i, i = 1, \ldots, N$, are unknown and, to gather information, a group of n of the N people is "randomly chosen" in the sense that all of the $\binom{N}{n}$ subsets of size n are equally likely to be chosen. These n people are then questioned and their feelings determined. If S denotes the sum of the n sampled values, determine its mean and variance.

An important application of the preceding problem is to a forthcoming election in which each person in the population is either for or against a certain candidate or proposition. If we take v_i to equal 1 if person i is in favor and 0 if he or she is against, then $\bar{v} = \sum_{i=1}^{N} v_i/N$ represents the proportion of the population that is in favor. To estimate $\bar{v}$, a random sample of n people is chosen, and these people are polled. The proportion of those polled who are in favor—that is, S/n—is often used as an estimate of $\bar{v}$.

Solution. For each person $i, i = 1, \ldots, N$, define an indicator variable I_i to indicate whether or not that person is included in the sample. That is,

$$I_i = \begin{cases} 1 & \text{if person } i \text{ is in the random sample} \\ 0 & \text{otherwise} \end{cases}$$

Now, S can be expressed by

$$S = \sum_{i=1}^{N} v_i I_i$$

so

$$E[S] = \sum_{i=1}^{N} v_i E[I_i]$$

$$\begin{aligned}\text{Var}(S) &= \sum_{i=1}^{N} \text{Var}(v_i I_i) + 2\sum\sum_{i<j} \text{Cov}(v_i I_i, v_j I_j) \\ &= \sum_{i=1}^{N} v_i^2 \text{Var}(I_i) + 2\sum\sum_{i<j} v_i v_j \text{Cov}(I_i, I_j)\end{aligned}$$

Because

$$\begin{aligned}E[I_i] &= \frac{n}{N} \\ E[I_i I_j] &= \frac{n}{N}\frac{n-1}{N-1}\end{aligned}$$

it follows that

$$\begin{aligned}\text{Var}(I_i) &= \frac{n}{N}\left(1 - \frac{n}{N}\right) \\ \text{Cov}(I_i, I_j) &= \frac{n(n-1)}{N(N-1)} - \left(\frac{n}{N}\right)^2 \\ &= \frac{-n(N-n)}{N^2(N-1)}\end{aligned}$$

Hence,

$$\begin{aligned}E[S] &= n\sum_{i=1}^{N} \frac{v_i}{N} = n\bar{v} \\ \text{Var}(S) &= \frac{n}{N}\left(\frac{N-n}{N}\right)\sum_{i=1}^{N} v_i^2 - \frac{2n(N-n)}{N^2(N-1)}\sum\sum_{i<j} v_i v_j\end{aligned}$$

The expression for $\text{Var}(S)$ can be simplified somewhat by using the identity $(v_1 + \cdots + v_N)^2 = \sum_{i=1}^{N} v_i^2 + 2\sum\sum_{i<j} v_i v_j$. After some simplification, we obtain

$$\text{Var}(S) = \frac{n(N-n)}{N-1}\left(\frac{\sum_{i=1}^{N} v_i^2}{N} - \bar{v}^2\right)$$

Consider now the special case in which Np of the v's are equal to 1 and the remainder equal to 0. Then, in this case, S is a hypergeometric random variable and has mean and variance given, respectively, by

$$E[S] = n\bar{v} = np \qquad \text{since } \bar{v} = \frac{Np}{N} = p$$

and

$$\begin{aligned}\text{Var}(S) &= \frac{n(N-n)}{N-1}\left(\frac{Np}{N} - p^2\right) \\ &= \frac{n(N-n)}{N-1}p(1-p)\end{aligned}$$

The quantity S/n, equal to the proportion of those sampled which have values equal to 1, is such that

$$E\left[\frac{S}{n}\right] = p$$
$$\text{Var}\left(\frac{S}{n}\right) = \frac{N - n}{n(N - 1)}p(1 - p)$$ ■

The correlation of two random variables X and Y, denoted by $\rho(X, Y)$, is defined, as long as $\text{Var}(X)\ \text{Var}(Y)$ is positive, by

$$\rho(X, Y) = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X)\text{Var}(Y)}}$$

It can be shown that

$$-1 \leq \rho(X, Y) \leq 1 \tag{4.2}$$

To prove Equation (4.2), suppose that X and Y have variances given by σ_x^2 and σ_y^2, respectively. Then, on the one hand,

$$\begin{aligned} 0 &\leq \text{Var}\left(\frac{X}{\sigma_x} + \frac{Y}{\sigma_y}\right) \\ &= \frac{\text{Var}(X)}{\sigma_x^2} + \frac{\text{Var}(Y)}{\sigma_y^2} + \frac{2\text{Cov}(X, Y)}{\sigma_x\sigma_y} \\ &= 2[1 + \rho(X, Y)] \end{aligned}$$

implying that

$$-1 \leq \rho(X, Y)$$

On the other hand,

$$\begin{aligned} 0 &\leq \text{Var}\left(\frac{X}{\sigma_x} - \frac{Y}{\sigma_y}\right) \\ &= \frac{\text{Var}(X)}{\sigma_x^2} + \frac{\text{Var}Y}{(-\sigma_y)^2} - \frac{2\text{Cov}(X, Y)}{\sigma_x\sigma_y} \\ &= 2[1 - \rho(X, Y)] \end{aligned}$$

implying that

$$\rho(X, Y) \leq 1$$

which completes the proof of Equation (4.2).

In fact, since $\text{Var}(Z) = 0$ implies that Z is constant with probability 1 (this intuitive relationship will be rigorously proven in Chapter 8), it follows from the proof of Equation (4.2) that $\rho(X, Y) = 1$ implies that $Y = a + bX$, where $b = \sigma_y/\sigma_x > 0$ and $\rho(X, Y) = -1$ implies that $Y = a + bX$, where $b = -\sigma_y/\sigma_x < 0$. We leave it as an exercise for the reader to show that the reverse is also true: that if $Y = a + bX$, then $\rho(X, Y)$ is either $+1$ or -1, depending on the sign of b.

The correlation coefficient is a measure of the degree of linearity between X and Y. A value of $\rho(X, Y)$ near $+1$ or -1 indicates a high degree of linearity between X and Y, whereas a value near 0 indicates that such linearity is absent. A positive

value of $\rho(X, Y)$ indicates that Y tends to increase when X does, whereas a negative value indicates that Y tends to decrease when X increases. If $\rho(X, Y) = 0$, then X and Y are said to be *uncorrelated*.

EXAMPLE 4d

Let I_A and I_B be indicator variables for the events A and B. That is,

$$I_A = \begin{cases} 1 & \text{if } A \text{ occurs} \\ 0 & \text{otherwise} \end{cases}$$
$$I_B = \begin{cases} 1 & \text{if } B \text{ occurs} \\ 0 & \text{otherwise} \end{cases}$$

Then

$$E[I_A] = P(A)$$
$$E[I_B] = P(B)$$
$$E[I_A I_B] = P(AB)$$

so

$$\begin{aligned} \text{Cov}(I_A, I_B) &= P(AB) - P(A)P(B) \\ &= P(B)[P(A|B) - P(A)] \end{aligned}$$

Thus, we obtain the quite intuitive result that the indicator variables for A and B are either positively correlated, uncorrelated, or negatively correlated, depending on whether $P(A|B)$ is, respectively, greater than, equal to, or less than $P(A)$. ■

Our next example shows that the sample mean and a deviation from the sample mean are uncorrelated.

EXAMPLE 4e

Let $X_1, \ldots, X_n$ be independent and identically distributed random variables having variance σ^2. Show that

$$\text{Cov}(X_i - \overline{X}, \overline{X}) = 0$$

Solution. We have

$$\begin{aligned} \text{Cov}(X_i - \overline{X}, \overline{X}) &= \text{Cov}(X_i, \overline{X}) - \text{Cov}(\overline{X}, \overline{X}) \\ &= \text{Cov}\left(X_i, \frac{1}{n}\sum_{j=1}^{n} X_j\right) - \text{Var}(\overline{X}) \\ &= \frac{1}{n}\sum_{j=1}^{n} \text{Cov}(X_i, X_j) - \frac{\sigma^2}{n} \\ &= \frac{\sigma^2}{n} - \frac{\sigma^2}{n} = 0 \end{aligned}$$

where the next-to-last equality uses the result of Example 4a and the final equality follows because

$$\text{Cov}(X_i, X_j) = \begin{cases} 0 & \text{if } j \neq i \text{ by independence} \\ \sigma^2 & \text{if } j = i \text{ since } \text{Var}(X_i) = \sigma^2 \end{cases}$$

Although $\overline{X}$ and the deviation $X_i - \overline{X}$ are uncorrelated, they are not, in general, independent. However, in the special case where the X_i are normal random variables, it turns out that not only is $\overline{X}$ independent of a single deviation, but it is independent of the entire sequence of deviations $X_j - \overline{X}, j = 1, \ldots, n$. This result will be established in Section 7.8, where we will also show that, in this case, the sample mean $\overline{X}$ and the sample variance S^2 are independent, with $(n-1)S^2/\sigma^2$ having a chi-squared distribution with $n-1$ degrees of freedom. (See Example 4a for the definition of S^2.) ■

EXAMPLE 4f

Consider m independent trials, each of which results in any of r possible outcomes with probabilities $P_1, P_2, \ldots, P_r, \sum_1^r P_i = 1$. If we let $N_i, i = 1, \ldots, r$, denote the number of the m trials that result in outcome i, then $N_1, N_2, \ldots, N_r$ have the multinomial distribution

$$P\{N_1 = n_1, N_2 = n_2, \ldots, N_r = n_r\} = \frac{m!}{n_1! n_2! \ldots n_r!} P_1^{n_1} P_2^{n_2} \cdots P_r^{n_r} \sum_{i=1}^r n_i = m$$

For $i \neq j$, it seems likely that when N_i is large, N_j would tend to be small; hence, it is intuitive that they should be negatively correlated. Let us compute their covariance by using Proposition 4.2(iv) and the representation

$$N_i = \sum_{k=1}^m I_i(k) \qquad \text{and} \qquad N_j = \sum_{k=1}^m I_j(k)$$

where

$$I_i(k) = \begin{cases} 1 & \text{if trial } k \text{ results in outcome } i \\ 0 & \text{otherwise} \end{cases}$$

$$I_j(k) = \begin{cases} 1 & \text{if trial } k \text{ results in outcome } j \\ 0 & \text{otherwise} \end{cases}$$

From Proposition 4.2(iv), we have

$$\text{Cov}(N_i, N_j) = \sum_{\ell=1}^m \sum_{k=1}^m \text{Cov}(I_i(k), I_j(\ell))$$

Now, on the one hand, when $k \neq \ell$,

$$\text{Cov}(I_i(k), I_j(\ell)) = 0$$

since the outcome of trial k is independent of the outcome of trial ℓ. On the other hand,

$$\begin{aligned}\text{Cov}(I_i(\ell), I_j(\ell)) &= E[I_i(\ell)I_j(\ell)] - E[I_i(\ell)]E[I_j(\ell)] \\ &= 0 - P_iP_j = -P_iP_j\end{aligned}$$

where the equation uses the fact that $I_i(\ell)I_j(\ell) = 0$, since trial ℓ cannot result in both outcome i and outcome j. Hence, we obtain

$$\text{Cov}(N_i, N_j) = -mP_iP_j$$

which is in accord with our intuition that N_i and N_j are negatively correlated. ■

7.5 CONDITIONAL EXPECTATION

7.5.1 Definitions

Recall that if X and Y are jointly discrete random variables, then the conditional probability mass function of X, given that $Y = y$, is defined, for all y such that $P\{Y = y\} > 0$, by

$$p_{X|Y}(x|y) = P\{X = x|Y = y\} = \frac{p(x, y)}{p_Y(y)}$$

It is therefore natural to define, in this case, the conditional expectation of X given that $Y = y$, for all values of y such that $p_Y(y) > 0$, by

$$\begin{aligned}E[X|Y = y] &= \sum_x xP\{X = x|Y = y\} \\ &= \sum_x xp_{X|Y}(x|y)\end{aligned}$$

EXAMPLE 5a

If X and Y are independent binomial random variables with identical parameters n and p, calculate the conditional expected value of X given that $X + Y = m$.

Solution. Let us first calculate the conditional probability mass function of X given that $X + Y = m$. For $k \le \min(n, m)$,

$$\begin{aligned}P\{X = k|X + Y = m\} &= \frac{P\{X = k, X + Y = m\}}{P\{X + Y = m\}} \\ &= \frac{P\{X = k, Y = m - k\}}{P\{X + Y = m\}} \\ &= \frac{P\{X = k\}P\{Y = m - k\}}{P\{X + Y = m\}} \\ &= \frac{\binom{n}{k}p^k(1-p)^{n-k}\binom{n}{m-k}p^{m-k}(1-p)^{n-m+k}}{\binom{2n}{m}p^m(1-p)^{2n-m}} \\ &= \frac{\binom{n}{k}\binom{n}{m-k}}{\binom{2n}{m}}\end{aligned}$$

7.7 MOMENT GENERATING FUNCTIONS

The moment generating function $M(t)$ of the random variable X is defined for all real values of t by

$$M(t) = E[e^{tX}]$$
$$= \begin{cases} \sum_x e^{tx} p(x) & \text{if } X \text{ is discrete with mass function } p(x) \\ \int_{-\infty}^{\infty} e^{tx} f(x)\, dx & \text{if } X \text{ is continuous with density } f(x) \end{cases}$$

We call $M(t)$ the moment generating function because all of the moments of X can be obtained by successively differentiating $M(t)$ and then evaluating the result at $t = 0$. For example,

$$\begin{aligned} M'(t) &= \frac{d}{dt} E[e^{tX}] \\ &= E\left[\frac{d}{dt}(e^{tX})\right] \\ &= E[Xe^{tX}] \end{aligned} \tag{7.1}$$

where we have assumed that the interchange of the differentiation and expectation operators is legitimate. That is, we have assumed that

$$\frac{d}{dt}\left[\sum_x e^{tx} p(x)\right] = \sum_x \frac{d}{dt}[e^{tx} p(x)]$$

in the discrete case and

$$\frac{d}{dt}\left[\int e^{tx} f(x)\, dx\right] = \int \frac{d}{dt}[e^{tx} f(x)]\, dx$$

in the continuous case. This assumption can almost always be justified and, indeed, is valid for all of the distributions considered in this book. Hence, from Equation (7.1), evaluated at $t = 0$, we obtain

$$M'(0) = E[X]$$

Similarly,

$$\begin{aligned} M''(t) &= \frac{d}{dt} M'(t) \\ &= \frac{d}{dt} E[Xe^{tX}] \\ &= E\left[\frac{d}{dt}(Xe^{tX})\right] \\ &= E[X^2 e^{tX}] \end{aligned}$$

Thus,

$$M''(0) = E[X^2]$$

In general, the nth derivative of $M(t)$ is given by

$$M^n(t) = E[X^n e^{tX}] \quad n \geq 1$$

implying that

$$M^n(0) = E[X^n] \quad n \geq 1$$

We now compute $M(t)$ for some common distributions.

EXAMPLE 7a Binomial distribution with parameters n and p

If X is a binomial random variable with parameters n and p, then

$$\begin{aligned} M(t) &= E[e^{tX}] \\ &= \sum_{k=0}^{n} e^{tk} \binom{n}{k} p^k (1 - p)^{n-k} \\ &= \sum_{k=0}^{n} \binom{n}{k} (pe^t)^k (1 - p)^{n-k} \\ &= (pe^t + 1 - p)^n \end{aligned}$$

where the last equality follows from the binomial theorem. Differentiation yields

$$M'(t) = n(pe^t + 1 - p)^{n-1} pe^t$$

Thus,

$$E[X] = M'(0) = np$$

Differentiating a second time yields

$$M''(t) = n(n - 1)(pe^t + 1 - p)^{n-2}(pe^t)^2 + n(pe^t + 1 - p)^{n-1} pe^t$$

so

$$E[X^2] = M''(0) = n(n - 1)p^2 + np$$

The variance of X is given by

$$\begin{aligned} \text{Var}(X) &= E[X^2] - (E[X])^2 \\ &= n(n - 1)p^2 + np - n^2p^2 \\ &= np(1 - p) \end{aligned}$$

verifying the result obtained previously. ■

EXAMPLE 7b Poisson distribution with mean λ

If X is a Poisson random variable with parameter λ, then

$$\begin{aligned} M(t) &= E[e^{tX}] \\ &= \sum_{n=0}^{\infty} \frac{e^{tn} e^{-\lambda} \lambda^n}{n!} \\ &= e^{-\lambda} \sum_{n=0}^{\infty} \frac{(\lambda e^t)^n}{n!} \end{aligned}$$

$$= e^{-\lambda} e^{\lambda e^t}$$
$$= \exp\{\lambda(e^t - 1)\}$$

Differentiation yields

$$M'(t) = \lambda e^t \exp\{\lambda(e^t - 1)\}$$
$$M''(t) = (\lambda e^t)^2 \exp\{\lambda(e^t - 1)\} + \lambda e^t \exp\{\lambda(e^t - 1)\}$$

Thus,

$$E[X] = M'(0) = \lambda$$
$$E[X^2] = M''(0) = \lambda^2 + \lambda$$
$$\text{Var}(X) = E[X^2] - (E[X])^2$$
$$= \lambda$$

Hence, both the mean and the variance of the Poisson random variable equal λ. ■

EXAMPLE 7c Exponential distribution with parameter λ

$$M(t) = E[e^{tX}]$$
$$= \int_0^\infty e^{tx} \lambda e^{-\lambda x}\, dx$$
$$= \lambda \int_0^\infty e^{-(\lambda - t)x}\, dx$$
$$= \frac{\lambda}{\lambda - t} \qquad \text{for } t < \lambda$$

We note from this derivation that, for the exponential distribution, $M(t)$ is defined only for values of t less than λ. Differentiation of $M(t)$ yields

$$M'(t) = \frac{\lambda}{(\lambda - t)^2} \quad M''(t) = \frac{2\lambda}{(\lambda - t)^3}$$

Hence,

$$E[X] = M'(0) = \frac{1}{\lambda} \quad E[X^2] = M''(0) = \frac{2}{\lambda^2}$$

The variance of X is given by

$$\text{Var}(X) = E[X^2] - (E[X])^2$$
$$= \frac{1}{\lambda^2}$$

■

EXAMPLE 7d Normal distribution

We first compute the moment generating function of a unit normal random variable with parameters 0 and 1. Letting Z be such a random variable, we have

$$M_Z(t) = E[e^{tZ}]$$
$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^\infty e^{tx} e^{-x^2/2}\, dx$$

$$
\begin{aligned}
&= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \exp\left\{-\frac{(x^2 - 2tx)}{2}\right\} dx \\
&= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \exp\left\{-\frac{(x - t)^2}{2} + \frac{t^2}{2}\right\} dx \\
&= e^{t^2/2} \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-(x-t)^2/2}\, dx \\
&= e^{t^2/2}
\end{aligned}
$$

Hence, the moment generating function of the unit normal random variable Z is given by $M_Z(t) = e^{t^2/2}$. To obtain the moment generating function of an arbitrary normal random variable, we recall (see Section 5.4) that $X = \mu + \sigma Z$ will have a normal distribution with parameters μ and σ^2 whenever Z is a unit normal random variable. Hence, the moment generating function of such a random variable is given by

$$
\begin{aligned}
M_X(t) &= E[e^{tX}] \\
&= E[e^{t(\mu+\sigma Z)}] \\
&= E[e^{t\mu} e^{t\sigma Z}] \\
&= e^{t\mu} E[e^{t\sigma Z}] \\
&= e^{t\mu} M_Z(t\sigma) \\
&= e^{t\mu} e^{(t\sigma)^2/2} \\
&= \exp\left\{\frac{\sigma^2 t^2}{2} + \mu t\right\}
\end{aligned}
$$

By differentiating, we obtain

$$
\begin{aligned}
M'_X(t) &= (\mu + t\sigma^2) \exp\left\{\frac{\sigma^2 t^2}{2} + \mu t\right\} \\
M''_X(t) &= (\mu + t\sigma^2)^2 \exp\left\{\frac{\sigma^2 t^2}{2} + \mu t\right\} + \sigma^2 \exp\left\{\frac{\sigma^2 t^2}{2} + \mu t\right\}
\end{aligned}
$$

Thus,

$$
\begin{aligned}
E[X] &= M'(0) = \mu \\
E[X^2] &= M''(0) = \mu^2 + \sigma^2
\end{aligned}
$$

implying that

$$
\begin{aligned}
\mathrm{Var}(X) &= E[X^2] - E([X])^2 \\
&= \sigma^2
\end{aligned}
$$

■

Tables 7.1 and 7.2 give the moment generating functions for some common discrete and continuous distributions.

An important property of moment generating functions is that the moment generating function of the sum of independent random variables equals the product of the individual moment generating functions. To prove this, suppose that X and Y are

TABLE 7.1: DISCRETE PROBABILITY DISTRIBUTION

	Probability mass function, $p(x)$	Moment generating function, $M(t)$	Mean	Variance
Binomial with parameters n, p; $0 \le p \le 1$	$\binom{n}{x} p^x(1-p)^{n-x}$ $x = 0, 1, \ldots, n$	$(pe^t + 1 - p)^n$	np	$np(1-p)$
Poisson with parameter $\lambda > 0$	$e^{-\lambda}\frac{\lambda^x}{x!}$ $x = 0, 1, 2, \ldots$	$\exp\{\lambda(e^t - 1)\}$	λ	λ
Geometric with parameter $0 \le p \le 1$	$p(1-p)^{x-1}$ $x = 1, 2, \ldots$	$\frac{pe^t}{1-(1-p)e^t}$	$\frac{1}{p}$	$\frac{1-p}{p^2}$
Negative binomial with parameters r, p; $0 \le p \le 1$	$\binom{n-1}{r-1} p^r(1-p)^{n-r}$ $n = r, r+1, \ldots$	$\left[\frac{pe^t}{1-(1-p)e^t}\right]^r$	$\frac{r}{p}$	$\frac{r(1-p)}{p^2}$

independent and have moment generating functions $M_X(t)$ and $M_Y(t)$, respectively. Then $M_{X+Y}(t)$, the moment generating function of $X + Y$, is given by

$$\begin{aligned} M_{X+Y}(t) &= E[e^{t(X+Y)}] \\ &= E[e^{tX}e^{tY}] \\ &= E[e^{tX}]E[e^{tY}] \\ &= M_X(t)M_Y(t) \end{aligned}$$

where the next-to-last equality follows from Proposition 4.1, since X and Y are independent.

Another important result is that the moment generating function uniquely determines the distribution. That is, if $M_X(t)$ exists and is finite in some region about $t = 0$, then the distribution of X is uniquely determined. For instance, if

$$M_X(t) = \left(\frac{1}{2}\right)^{10}(e^t + 1)^{10},$$

then it follows from Table 7.1 that X is a binomial random variable with parameters 10 and $\frac{1}{2}$.

EXAMPLE 7e

Suppose that the moment generating function of a random variable X is given by $M(t) = e^{3(e^t-1)}$. What is $P\{X = 0\}$?

TABLE 7.2: CONTINUOUS PROBABILITY DISTRIBUTION

	Probability mass function, $f(x)$	Moment generating function, $M(t)$	Mean	Variance
Uniform over (a, b)	$f(x) = \begin{cases} \dfrac{1}{b-a} & a < x < b \\ 0 & \text{otherwise} \end{cases}$	$\dfrac{e^{tb} - e^{ta}}{t(b-a)}$	$\dfrac{a+b}{2}$	$\dfrac{(b-a)^2}{12}$
Exponential with parameter $\lambda > 0$	$f(x) = \begin{cases} \lambda e^{-\lambda x} & x \geq 0 \\ 0 & x < 0 \end{cases}$	$\dfrac{\lambda}{\lambda - t}$	$\dfrac{1}{\lambda}$	$\dfrac{1}{\lambda^2}$
Gamma with parameters $(s, \lambda), \lambda > 0$	$f(x) = \begin{cases} \dfrac{\lambda e^{-\lambda x}(\lambda x)^{s-1}}{\Gamma(s)} & x \geq 0 \\ 0 & x < 0 \end{cases}$	$\left(\dfrac{\lambda}{\lambda - t}\right)^s$	$\dfrac{s}{\lambda}$	$\dfrac{s}{\lambda^2}$
Normal with parameters (μ, σ^2)	$f(x) = \dfrac{1}{\sqrt{2\pi}\sigma} e^{-(x-\mu)^2/2\sigma^2} \quad -\infty < x < \infty$	$\exp\left\{\mu t + \dfrac{\sigma^2 t^2}{2}\right\}$	μ	σ^2

Solution. We see from Table 7.1 that $M(t) = e^{3(e^t-1)}$ is the moment generating function of a Poisson random variable with mean 3. Hence, by the one-to-one correspondence between moment generating functions and distribution functions, it follows that X must be a Poisson random variable with mean 3. Thus, $P\{X = 0\} = e^{-3}$. ■

EXAMPLE 7f Sums of independent binomial random variables

If X and Y are independent binomial random variables with parameters (n, p) and (m, p), respectively, what is the distribution of $X + Y$?

Solution. The moment generating function of $X + Y$ is given by

$$\begin{aligned} M_{X+Y}(t) = M_X(t)M_Y(t) &= (pe^t + 1 - p)^n(pe^t + 1 - p)^m \\ &= (pe^t + 1 - p)^{m+n} \end{aligned}$$

However, $(pe^t + 1 - p)^{m+n}$ is the moment generating function of a binomial random variable having parameters $m + n$ and p. Thus, this must be the distribution of $X + Y$. ■

EXAMPLE 7g Sums of independent Poisson random variables

Calculate the distribution of $X + Y$ when X and Y are independent Poisson random variables with means respective λ_1 and λ_2.

Solution.

$$\begin{aligned} M_{X+Y}(t) &= M_X(t)M_Y(t) \\ &= \exp\{\lambda_1(e^t - 1)\}\exp\{\lambda_2(e^t - 1)\} \\ &= \exp\{(\lambda_1 + \lambda_2)(e^t - 1)\} \end{aligned}$$

Hence, $X + Y$ is Poisson distributed with mean $\lambda_1 + \lambda_2$, verifying the result given in Example 3e of Chapter 6. ■

EXAMPLE 7h Sums of independent normal random variables

Show that if X and Y are independent normal random variables with respective parameters (μ_1, σ_1^2) and (μ_2, σ_2^2), then $X + Y$ is normal with mean $\mu_1 + \mu_2$ and variance $\sigma_1^2 + \sigma_2^2$.

Solution.

$$\begin{aligned} M_{X+Y}(t) &= M_X(t)M_Y(t) \\ &= \exp\left\{\frac{\sigma_1^2 t^2}{2} + \mu_1 t\right\}\exp\left\{\frac{\sigma_2^2 t^2}{2} + \mu_2 t\right\} \\ &= \exp\left\{\frac{(\sigma_1^2 + \sigma_2^2)t^2}{2} + (\mu_1 + \mu_2)t\right\} \end{aligned}$$

which is the moment generating function of a normal random variable with mean $\mu_1 + \mu_2$ and variance $\sigma_1^2 + \sigma_2^2$. The desired result then follows because the moment generating function uniquely determines the distribution. ■

EXAMPLE 7i

Compute the moment generating function of a chi-squared random variable with n degrees of freedom.

Solution. We can represent such a random variable as

$$Z_1^2 + \cdots + Z_n^2$$

where $Z_1, \ldots, Z_n$ are independent standard normal random variables. Let $M(t)$ be its moment generating function. Then, by the preceding,

$$M(t) = (E[e^{tZ^2}])^n$$

where Z is a standard normal random variable. Now,

$$\begin{aligned} E[e^{tZ^2}] &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{tx^2} e^{-x^2/2}\, dx \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-x^2/2\sigma^2}\, dx \quad \text{where } \sigma^2 = (1 - 2t)^{-1} \\ &= \sigma \\ &= (1 - 2t)^{-1/2} \end{aligned}$$

where the next-to-last equality uses the fact that the normal density with mean 0 and variance σ^2 integrates to 1. Therefore,

$$M(t) = (1 - 2t)^{-n/2}$$ ■

EXAMPLE 7j Moment generating function of the sum of a random number of random variables

Let $X_1, X_2, \ldots$ be a sequence of independent and identically distributed random variables, and let N be a nonnegative, integer-valued random variable that is independent of the sequence $X, i \geq 1$. We want to compute the moment generating function of

$$Y = \sum_{i=1}^{N} X_i$$

(In Example 5d, Y was interpreted as the amount of money spent in a store on a given day when both the amount spent by a customer and the number of customers are random variables.)

To compute the moment generating function of Y, we first condition on N as follows:

$$\begin{aligned} E\left[\exp\left\{ t \sum_1^N X_i \right\} \middle| N = n \right] &= E\left[\exp\left\{ t \sum_1^n X_i \right\} \middle| N = n \right] \\ &= E\left[\exp\left\{ t \sum_1^n X_i \right\} \right] \\ &= [M_X(t)]^n \end{aligned}$$

where

$$M_X(t) = E[e^{tX_i}]$$

Hence,

$$E[e^{tY}|N] = (M_X(t))^N$$

Thus,

$$M_Y(t) = E[(M_X(t))^N]$$

The moments of Y can now be obtained upon differentiation, as follows:

$$M'_Y(t) = E[N(M_X(t))^{N-1}M'_X(t)]$$

So

$$\begin{aligned} E[Y] &= M'_Y(0) \\ &= E[N(M_X(0))^{N-1}M'_X(0)] \\ &= E[NEX] \\ &= E[N]E[X] \end{aligned} \tag{7.2}$$

verifying the result of Example 5d. (In this last set of equalities, we have used the fact that $M_X(0) = E[e^{0X}] = 1$.)

Also,

$$M''_Y(t) = E[N(N - 1)(M_X(t))^{N-2}(M'_X(t))^2 + N(M_X(t))^{N-1}M''_X(t)]$$

so

$$\begin{aligned} E[Y^2] &= M''_Y(0) \\ &= E[N(N - 1)(E[X])^2 + NE[X^2]] \\ &= (E[X])^2(E[N^2] - E[N]) + E[N]E[X^2] \\ &= E[N](E[X^2] - (E[X])^2) + (E[X])^2E[N^2] \\ &= E[N]\text{Var}(X) + (E[X])^2E[N^2] \end{aligned} \tag{7.3}$$

Hence, from Equations (7.2) and (7.3), we have

$$\begin{aligned} \text{Var}(Y) &= E[N]\text{Var}(X) + (E[X])^2(E[N^2] - (E[N])^2) \\ &= E[N]\text{Var}(X) + (E[X])^2\text{Var}(N) \end{aligned}$$

■

EXAMPLE 7k

Let Y denote a uniform random variable on (0, 1), and suppose that, conditional on $Y = p$, the random variable X has a binomial distribution with parameters n and p. In Example 5k, we showed that X is equally likely to take on any of the values $0, 1, \ldots, n$. Establish this result by using moment generating functions.

Solution. To compute the moment generating function of X, start by conditioning on the value of Y. Using the formula for the binomial moment generating function gives

$$E[e^{tX}|Y = p] = (pe^t + 1 - p)^n$$

Now, Y is uniform on $(0, 1)$, so, upon taking expectations, we obtain

$$\begin{aligned} E[e^{tX}] &= \int_0^1 (pe^t + 1 - p)^n \, dp \\ &= \frac{1}{e^t - 1} \int_1^{e^t} y^n dy \quad \text{(by the substitution } y = pe^t + 1 - p) \\ &= \frac{1}{n+1} \frac{e^{t(n+1)} - 1}{e^t - 1} \\ &= \frac{1}{n+1}(1 + e^t + e^{2t} + \cdots + e^{nt}) \end{aligned}$$

Because the preceding is the moment generating function of a random variable that is equally likely to be any of the values $0, 1, \ldots, n$, the desired result follows from the fact that the moment generating function of a random variable uniquely determines its distribution. ■

7.7.1 Joint Moment Generating Functions

It is also possible to define the joint moment generating function of two or more random variables. This is done as follows: For any n random variables $X_1, \ldots, X_n$, the joint moment generating function, $M(t_1, \ldots, t_n)$, is defined, for all real values of $t_1, \ldots, t_n$, by

$$M(t_1, \ldots, t_n) = E[e^{t_1 X_1 + \cdots + t_n X_n}]$$

The individual moment generating functions can be obtained from $M(t_1, \ldots, t_n)$ by letting all but one of the t_j's be 0. That is,

$$M_{X_i}(t) = E[e^{tX_i}] = M(0, \ldots, 0, t, 0, \ldots, 0)$$

where the t is in the ith place.

It can be proven (although the proof is too advanced for this text) that the joint moment generating function $M(t_1, \ldots, t_n)$ uniquely determines the joint distribution of $X_1, \ldots, X_n$. This result can then be used to prove that the n random variables $X_1, \ldots, X_n$ are independent if and only if

$$M(t_1, \ldots, t_n) = M_{X_1}(t_1) \cdots M_{X_n}(t_n) \tag{7.4}$$

For the proof in one direction, if the n random variables are independent, then

$$\begin{aligned} M(t_1, \ldots, t_n) &= E[e^{(t_1 X_1 + \cdots + t_n X_n)}] \\ &= E[e^{t_1 X_1} \cdots e^{t_n X_n}] \\ &= E[e^{t_1 X_1}] \cdots E[e^{t_n X_n}] \quad \text{by independence} \\ &= M_{X_1}(t_1) \cdots M_{X_n}(t_n) \end{aligned}$$

For the proof in the other direction, if Equation (7.4) is satisfied, then the joint moment generating function $M(t_1, \ldots, t_n)$ is the same as the joint moment generating function of n independent random variables, the ith of which has the same distribution as X_i. As the joint moment generating function uniquely determines the joint distribution, this must be the joint distribution; hence, the random variables are independent.

EXAMPLE 7l

Let X and Y be independent normal random variables, each with mean μ and variance σ^2. In Example 7a of Chapter 6, we showed that $X + Y$ and $X - Y$ are independent. Let us now establish this result by computing their joint moment generating function:

$$\begin{aligned} E[e^{t(X+Y)+s(X-Y)}] &= E[e^{(t+s)X+(t-s)Y}] \\ &= E[e^{(t+s)X}]E[e^{(t-s)Y}] \\ &= e^{\mu(t+s)+\sigma^2(t+s)^2/2}e^{\mu(t-s)+\sigma^2(t-s)^2/2} \\ &= e^{2\mu t+\sigma^2 t^2}e^{\sigma^2 s^2} \end{aligned}$$

But we recognize the preceding as the joint moment generating function of the sum of a normal random variable with mean 2μ and variance $2\sigma^2$ and an independent normal random variable with mean 0 and variance $2\sigma^2$. Because the joint moment generating function uniquely determines the joint distribution, it follows that $X + Y$ and $X - Y$ are independent normal random variables. ■

In the next example, we use the joint moment generating function to verify a result that was established in Example 2b of Chapter 6.

EXAMPLE 7m

Suppose that the number of events that occur is a Poisson random variable with mean λ and that each event is independently counted with probability p. Show that the number of counted events and the number of uncounted events are independent Poisson random variables with respective means λp and $\lambda(1 - p)$.

Solution. Let X denote the total number of events, and let X_c denote the number of them that are counted. To compute the joint moment generating function of X_c, the number of events that are counted, and $X - X_c$, the number that are uncounted, start by conditioning on X to obtain

$$\begin{aligned} E[e^{sX_c+t(X-X_c)}|X = n] &= e^{tn}E[e^{(s-t)X_c}|X = n] \\ &= e^{tn}(pe^{s-t} + 1 - p)^n \\ &= (pe^s + (1 - p)e^t)^n \end{aligned}$$

which follows because, conditional on $X = n, X_c$ is a binomial random variable with parameters n and p. Hence,

$$E[e^{sX_c+t(X-X_c)}|X] = (pe^s + (1 - p)e^t)^X$$

Taking expectations of both sides of this equation yields

$$E[e^{sX_c+t(X-X_c)}] = E[(pe^s + (1 - p)e^t)^X]$$

Now, since X is Poisson with mean λ, it follows that $E[e^{tX}] = e^{\lambda(e^t-1)}$. Therefore, for any positive value a we see (by letting $a = e^t$) that $E[a^X] = e^{\lambda(a-1)}$. Thus

$$\begin{aligned} E[e^{sX_c+t(X-X_c)}] &= e^{\lambda(pe^s+(1-p)e^t-1)} \\ &= e^{\lambda p(e^s-1)}e^{\lambda(1-p)(e^t-1)} \end{aligned}$$

As the preceding is the joint moment generating function of independent Poisson random variables with respective means λp and $\lambda(1 - p)$, the result is proven. ■

7.8 ADDITIONAL PROPERTIES OF NORMAL RANDOM VARIABLES

7.8.1 The Multivariate Normal Distribution

Let $Z_1, \ldots, Z_n$ be a set of n independent unit normal random variables. If, for some constants $a_{ij}, 1 \le i \le m, 1 \le j \le n$, and $\mu_i, 1 \le i \le m$,

$$\begin{aligned} X_1 &= a_{11}Z_1 + \cdots + a_{1n}Z_n + \mu_1 \\ X_2 &= a_{21}Z_1 + \cdots + a_{2n}Z_n + \mu_2 \\ &\vdots \\ X_i &= a_{i1}Z_1 + \cdots + a_{in}Z_n + \mu_i \\ &\vdots \\ X_m &= a_{m1}Z_1 + \cdots + a_{mn}Z_n + \mu_m \end{aligned}$$

then the random variables $X_1, \ldots, X_m$ are said to have a multivariate normal distribution.

From the fact that the sum of independent normal random variables is itself a normal random variable, it follows that each X_i is a normal random variable with mean and variance given, respectively, by

$$\begin{aligned} E[X_i] &= \mu_i \\ \mathrm{Var}(X_i) &= \sum_{j=1}^{n} a_{ij}^2 \end{aligned}$$

Let us now consider

$$M(t_1, \ldots, t_m) = E[\exp\{t_1X_1 + \cdots + t_mX_m\}]$$

the joint moment generating function of $X_1, \ldots, X_m$. The first thing to note is that since $\sum_{i=1}^{m} t_iX_i$ is itself a linear combination of the independent normal random variables $Z_1, \ldots, Z_n$, it is also normally distributed. Its mean and variance are

$$E\left[\sum_{i=1}^{m} t_iX_i\right] = \sum_{i=1}^{m} t_i\mu_i$$

and

$$\begin{aligned} \mathrm{Var}\left(\sum_{i=1}^{m} t_iX_i\right) &= \mathrm{Cov}\left(\sum_{i=1}^{m} t_iX_i, \sum_{j=1}^{m} t_jX_j\right) \\ &= \sum_{i=1}^{m}\sum_{j=1}^{m} t_it_j\mathrm{Cov}(X_i, X_j) \end{aligned}$$